

NO-DORK: Numerically optimized dynamically orthogonal Runge-Kutta schemes

Sanaa Mouzahir

Center for Computational Science and Engineering
Massachusetts Institute of Technology
Cambridge, MA 02139, USA
sanaamz@mit.edu

Pierre F. J. Lermusiaux

Center for Computational Science and Engineering
Massachusetts Institute of Technology
Cambridge, MA 02139, USA

January 2026

Abstract

Large-scale simulations are constrained by computational cost and memory requirements, particularly when uncertainty quantification requires repeated solutions of time-dependent partial differential equations. Reduced-order models address this cost by restricting the state to a low-dimensional space. Dynamical low-rank approximation and dynamically orthogonal (DO) methods evolve that space, while rank-adaptive dynamically orthogonal Runge–Kutta (DORK) schemes also adjust its dimension. However, DORK advances every retained basis vector and coefficient at each time step, even when these factors evolve on different time scales. We introduce numerically optimized dynamically orthogonal Runge–Kutta (NO-DORK) schemes that jointly select the time step, rank, and subsets of basis vectors and coefficients to update. A constrained Dirac–Frenkel principle defines the selective dynamics, and a deflation formulation permits the use of existing DORK retractions. In the Karhunen–Loève gauge, the projection defect splits orthogonally into rank-truncation, basis-staleness, and coefficient-staleness components. The resulting scores determine optimal active sets for prescribed cardinalities. Local and global error bounds, combined with an operation-count model, define a cost–error Pareto problem for adaptive scheduling. Stochastic viscous Burgers experiments verify the defect identities and demonstrate the benefit of score-based selection. For the reported configuration, NO-DORK meets the prescribed error budget at 20% lower modeled cost than full-update rank-adaptive DORK with the same rank decisions and at 7% of the Monte Carlo cost.

Keywords. dynamical low-rank approximation; dynamically orthogonal equations; rank adaptivity; selective (partial) updates; perturbative retractions; Runge–Kutta; Pareto optimization; adaptive time stepping; uncertainty quantification; compressible Navier–Stokes.

MSC 2020. 65L05, 65L06, 65M12, 65M22, 65F55, 60H35, 76N06, 76M35, 90C29.

1 Introduction

Uncertainty quantification (UQ) and reduced-order modeling of time-dependent partial differential equations often involve a state matrix $X(t) \in \mathbb{R}^{m \times n}$. Its columns may represent realizations or parameters, while its rows represent spatial degrees of freedom. Monte Carlo sampling advances all n realizations and has a sampling error of order $n^{-1/2}$ [9]. Dynamical low-rank approximation (DLRA) [48] and the dynamically orthogonal (DO) field equations [68] instead evolve $\tilde{X} = UZ^\top$ on the manifold of rank- r matrices. Here $U \in \mathbb{R}^{m \times r}$ has orthonormal columns and $Z \in \mathbb{R}^{n \times r}$. Projecting the dynamics onto the tangent space yields the DO equations in the gauge $U^\top \dot{U} = 0$.

Low-rank evolution can reduce the cost substantially when $r \ll n$, although its benefit depends on the spectrum and the cost of evaluating the right-hand side.

Robust integration and rank adaptation have extended this approach. Projector splitting [51] and basis-update-and-Galerkin integrators [10] avoid the sensitivity of direct factor integration to small singular values. Dynamically orthogonal Runge–Kutta (DORK) schemes use perturbative retractions to update the subspace within a time step [13, 14]. Rank-adaptive methods change r as the effective dimension of the solution changes [11, 69].

These methods still advance all retained basis vectors and coefficients at every step. Yet the rotation rates of the modes and the rates of change of their coefficients need not be uniform. Selective updates can reduce work when some components change slowly. This observation is related to multirate integration [32, 67], but it requires an error measure tailored to the low-rank factors.

This work. We propose and analyze *selective-update* rank-adaptive DORK schemes. At each step, only an *active* set $\mathcal{A} \subseteq \{1, \dots, r\}$ of modes has its spatial basis vectors advanced, and only a set $\mathcal{B} \subseteq \{1, \dots, r\}$ of coefficient columns is advanced; the remaining basis vectors and coefficients are frozen. The sets $\mathcal{A}$ and $\mathcal{B}$, their cardinalities $a = |\mathcal{A}|$ and $b = |\mathcal{B}|$, the rank r and the time step Δt are chosen *jointly* at fixed scheduling instants by solving a multiobjective (cost-rate versus error-rate) Pareto problem using the error and cost models derived below. We analyze the effect of each frozen component and use its defect score to choose the active sets. Our contributions are:

1. **Selective DO equations and their geometry** (Section 4). We derive the selective dynamics from a constrained Dirac–Frenkel principle on the submanifold $\mathcal{M}_r^{\mathcal{F}} \subset \mathcal{M}_r$ of rank- r matrices whose column space contains the frozen basis $U_{\mathcal{F}}$, $\mathcal{F} = \{1, \dots, r\} \setminus \mathcal{A}$. The resulting projector splits orthogonally into a coefficient part and a basis part; the basis part is the DO mode equation restricted to the active columns but with the *active* covariance $Z_{\mathcal{A}}^{\top} Z_{\mathcal{A}}$, and it coincides with the corresponding columns of the full DO update exactly when the coefficients are uncorrelated (the Karhunen–Loève gauge).
2. **Deflation theorem and selective retractions** (Theorem 4.7). The selective DORK step equals the full DORK step of Charous and Lermusiaux [13, 14] applied to the *deflated* problem $(U_{\mathcal{A}} Z_{\mathcal{A}}^{\top}, P_{U_{\mathcal{F}}}^{\perp} \Phi)$, followed by the optimal (proper-orthogonal-decomposition-like) coefficient update $Z_{\mathcal{B}} \leftarrow \chi^{\top} U_{\mathcal{B}}$. Hence optimal perturbative retractions of arbitrary order, gradient-descent retractions, robust retractions and rank augmentation carry over verbatim, together with the stability bound $\|\tilde{X}_{i+1}\| \leq \|\tilde{X}_i + \Delta t \Phi\|$ whenever all coefficients are updated.
3. **Exact defect decomposition** (Theorem 4.4). In the Karhunen–Loève gauge, the squared local defect of the selective projection is

$$\left\| F - P_{\tilde{X}}^{\mathcal{A}, \mathcal{B}} F \right\|_{\text{F}}^2 = \underbrace{\left\| P_{U^{\perp}}^{\perp} F P_Z^{\perp} \right\|_{\text{F}}^2}_{\text{normal (truncation)}} + \underbrace{\sum_{j \notin \mathcal{A}} \sigma_j^2 \|\dot{u}_j\|^2}_{\text{basis staleness}} + \underbrace{\sum_{j \notin \mathcal{B}} \|\dot{z}_j\|^2}_{\text{coefficient staleness}},$$

where σ_j are the singular values of $\tilde{X}$, $\dot{u}_j$ the DO mode velocities and $\dot{z}_j$ the coefficient velocities. The price of freezing a mode is its amplitude times its rotation rate; the price of freezing a coefficient is its own rate of change. The decomposition is separable in j , so for fixed cardinalities the optimal $\mathcal{A}$ and $\mathcal{B}$ are found by sorting (Lemma 4.6).

4. **Error analysis** (Section 5). We prove a per-step bound in which the selective step differs from the full step by Δt times the staleness defects up to $\mathcal{O}(\Delta t^2)$ (Theorem 5.4), a global

Gronwall-type bound that combines the Runge–Kutta order p , the retraction order q , the normal defect and the two staleness defects with time-varying step sizes and active sets (Theorem 5.5), a bound on the loss of optimality caused by holding the selection fixed over a scheduling interval of length τ (Lemma 5.7), a certificate that allows frozen modes to be left unexamined until their score can have crossed the activation threshold (Corollary 5.8), and the interplay with rank augmentation and truncation (Proposition 5.9).

5. **Cost model and joint Pareto scheduling** (Section 6). We give an explicit per-step cost model $\mathbf{C}(\Delta t, r, a, b)$ for two evaluation strategies of the right-hand side (Galerkin contraction of polynomial nonlinearities and sampled/CUR evaluation of general nonlinearities), an error-rate model $\mathbf{E}(\Delta t, r, \mathcal{A}, \mathcal{B})$ assembled from computable quantities (embedded Runge–Kutta estimate, residual spectrum, staleness scores), and show that the multiobjective problem $\min(\mathbf{C}/\Delta t, \mathbf{E})$ reduces to an enumeration over $\mathcal{O}(r^2 n_r)$ closed-form candidates whose Pareto front is then reduced to a single decision by a knee or an error-budget rule.
6. **Verification** (Section 9.1). On a stochastic viscous Burgers equation with Monte Carlo references we confirm the defect identity and the local error theorem to rounding accuracy, show that score-based selection of the active set beats random selection and, by a factor of three to four in error at equal cost, the energy heuristic that keeps the most energetic modes active, and show that the joint scheduler meets a given error budget at 20% lower cost than full-update rank-adaptive DORK with the same rank decisions, at lower cost and error than fixed-rank DORK, and at 7% of the Monte Carlo cost.
7. **Application to stochastic three-dimensional compressible Navier–Stokes flows** (Sections 8 and 9.2). We specialize the formulation to the compressible Navier–Stokes equations (energy-weighted inner product, conservative variables, sampled evaluation of the non-polynomial fluxes) ; numerical assessment of this application remains to be completed.

Relation to fixed-basis reduced-order models. The limiting cases of the selective scheme are instructive. With $\mathcal{A} = \emptyset$ and $\mathcal{B} = \{1, \dots, r\}$, the basis is frozen and only the coefficients evolve through $\dot{\mathbf{Z}} = \mathbf{F}^\top \mathbf{U}$; this is a Galerkin reduced-order model on a fixed basis, as in proper orthogonal decomposition (POD) [52, 71], with the basis chosen as the current DO subspace rather than from offline snapshots. With $\mathcal{A} = \mathcal{B} = \{1, \dots, r\}$ the scheme is the rank-adaptive DORK method of Charous and Lermusiaux [14]. The staleness defects quantify exactly the error that online basis-adaptation methods [64, 65] set out to remove, and the scheduler chooses the point on this continuum where the marginal cost of rotating one more mode equals its marginal error.

Organization. Section 2 reviews the DLRA, DO, POD/reduced-order-modeling, sampling-based UQ and adaptive time-integration literatures. Section 3 recalls the DLRA/DO setting and the DORK schemes with perturbative retractions. Section 4 introduces the selective-update dynamics, the deflation theorem and selective retractions. Section 5 contains the error analysis. Section 6 develops the cost and error models and the joint Pareto scheduler; Section 7 collects the algorithms. Section 8 specializes the method to the stochastic compressible Navier–Stokes equations and Section 9 reports the numerical experiments. Section 10 concludes. Proofs are collected in Appendix A and notation in Appendix B.

2 Review of related frameworks

We place the selective scheme in the context of dynamical low-rank approximation, evolving stochastic bases, projection-based reduced-order models, uncertainty quantification, and adaptive integration. Table 1 compares the quantities that these approaches adapt.

2.1 Dynamical low-rank approximation

The Dirac–Frenkel variational principle selects the tangent velocity that best approximates the full dynamics [23, 31]. For matrix differential equations, DLRA constrains this velocity to the fixed-rank manifold [48]. Its tangent projector becomes sensitive to small singular values. Projector splitting and basis-update-and-Galerkin (BUG) integrators address this issue [10, 51]. Rank-adaptive BUG methods also enlarge or truncate the basis according to a tolerance [11].

DORK schemes combine Runge–Kutta stages with perturbative retractions and update the subspace within each step [13, 14]. The latter reference develops stable, optimal, gradient-descent, and robust retractions together with rank augmentation. Our selective basis update applies these constructions to a deflated active block. Rank indicators based on neglected dynamics or residuals supply complementary criteria for adding modes [42, 69]. The analysis below distinguishes rank truncation from the additional error caused by frozen modes and coefficients.

2.2 Dynamically orthogonal and evolving-basis methods

The DO equations evolve the mean, orthonormal spatial modes, and stochastic coefficients under the gauge $U^\top \dot{U} = 0$ [68]. Their relation to DLRA and their error properties have been studied for random PDEs [30, 59]. Adaptive DO methods add and remove modes as the effective stochastic dimension changes [69, 77]. Bi-orthogonal formulations evolve both spatial and stochastic bases [16, 17]. Time-dependent-basis methods further develop online reduced models, including sampled evaluation of nonlinear right-hand sides [24, 60]. These approaches motivate our compressible-flow formulation, for which direct evaluation of every realization is costly.

2.3 Proper orthogonal decomposition and reduced-order models

Proper orthogonal decomposition (POD) constructs a basis that captures ensemble or trajectory energy; Galerkin projection then evolves coefficients on that basis [44, 71]. Fixed bases can be inadequate for transport and changing dynamics, which motivates online subspace updates [64, 65]. The present method chooses which basis vectors to update at a given step. Compressible-flow reduction also requires an appropriate inner product and attention to stability [6, 66]. For non-polynomial right-hand sides, sampled interpolation and CUR approximations can reduce evaluation work [15, 24].

2.4 Uncertainty quantification and adaptive integration

Monte Carlo sampling provides a reference ensemble but requires many full solves for a small sampling error [9]. Polynomial chaos and stochastic collocation instead approximate dependence on random inputs in prescribed spaces [82, 83]. DO evolves a basis adapted to the instantaneous state. Its error consequently includes low-rank truncation in addition to any sampling or discretization error.

Embedded Runge–Kutta pairs control temporal error by adapting the step size [25, 40]. Multirate methods update slow components less frequently [32, 67]. Our method likewise reuses selected

Table 1: Quantities adapted in time by representative frameworks. Δt : time step; r : rank/number of modes; $\mathcal{A}$: set of modes whose basis is updated; $\mathcal{B}$: set of coefficients updated; $\mathcal{J}$: sampled realizations/points used to evaluate the right-hand side.

Framework	Δt	r	$\mathcal{A}$	$\mathcal{B}$	$\mathcal{J}$
Monte Carlo [55]	✓	–	–	–	–
POD–Galerkin ROM [44, 50]	✓	offline	$\emptyset$	all	(DEIM)
Online adaptive ROM [64, 65]	✓	fixed	low-rank update	all	✓
DO / DLRA [48, 68]	✓	fixed	all	all	–
Adaptive DO [69, 77]	✓	✓	all	all	–
Rank-adaptive BUG [11, 12]	✓	✓	all	all	–
TDB–CUR [24, 61]	✓	✓	all	all	✓
Rank-adaptive DORK [14]	✓	✓	all	all	–
Selective-update DORK (this work)	✓	✓	✓	✓	✓

factors, but bases its decisions on a decomposition of the low-rank projection defect. Cost–error Pareto selection then balances the available updates [8, 56].

2.5 Positioning

Table 1 makes the gap explicit. Existing adaptive low-rank schemes adapt the time step and the rank, and online ROMs adapt the sampling, but no framework adapts the *subset* of modes and coefficients that are advanced at a given step or optimizes that subset jointly with Δt and r against a cost–error trade-off. The selective-update DORK schemes fill this gap, and the error analysis of Section 5 certifies the resulting adaptive procedure.

3 Preliminaries: dynamical low-rank approximation and DORK schemes

3.1 Setting

We consider the matrix differential equation

$$\frac{dX}{dt} = F(X, t; \omega), \quad X(0) = X_0, \quad X(t) \in \mathbb{R}^{m \times n}, \quad (1)$$

where $\omega \in \Omega$ is an event of a probability space. In the UQ setting of Section 8, the j th column of X is the spatially discretized state of the j th realization (m spatial degrees of freedom, n realizations), and F is the spatially discretized right-hand side applied column-wise. Equation (1) also covers parametric problems, two-dimensional fields and second-order equations rewritten in first-order form [43, 48]. Unless stated otherwise, $\|\cdot\|$ denotes the Frobenius norm and $\langle A, B \rangle = \text{tr}(A^\top B)$. The weighted inner products needed for compressible flow are introduced in Section 8 and reduce to this case after a change of variables.

The manifold of rank- r matrices $\mathcal{M}_r = \{X \in \mathbb{R}^{m \times n} : \text{rank } X = r\}$ is a smooth embedded submanifold of dimension $(m + n - r)r$ [48, 78]. We parameterize its points as

$$\tilde{X} = UZ^\top, \quad U \in \mathcal{V}_{m,r} = \{U \in \mathbb{R}^{m \times r} : U^\top U = I_r\}, \quad Z \in \mathbb{R}_*^{n \times r}, \quad (2)$$

with $\mathbb{R}_*^{n \times r}$ the full-column-rank matrices, following the DO parameterization of Sapsis and Lermusiaux [68] as used in Charous and Lermusiaux [13], Feppon and Lermusiaux [30]. The coefficient

covariance is $C = Z^\top Z$, and the singular values $\sigma_1 \geq \dots \geq \sigma_r > 0$ of $\tilde{X}$ are the square roots of the eigenvalues of C . We say that the factorization is in the *Karhunen–Loève (KL) gauge* when $C = \Lambda = \text{diag}(\sigma_1^2, \dots, \sigma_r^2)$; any factorization can be brought to this gauge by the rotation $U \leftarrow UV$, $Z \leftarrow ZV$ with V the eigenvectors of C , which leaves $\tilde{X}$ unchanged.

3.2 Tangent-space projection and the DO equations

The tangent space of $\mathcal{M}_r$ at $\tilde{X} = UZ^\top$ and the orthogonal projector onto it are [48, Prop. 2.1]

$$T_{\tilde{X}}\mathcal{M}_r = \{\delta U Z^\top + U \delta Z^\top : \delta U \in \mathbb{R}^{m \times r}, \delta Z \in \mathbb{R}^{n \times r}\}, \quad P_{T_{\tilde{X}}\mathcal{M}_r}F = P_U F + P_U^\perp F P_Z, \quad (3)$$

with $P_U = UU^\top$, $P_U^\perp = I - P_U$ and $P_Z = Z(Z^\top Z)^{-1}Z^\top$. The DLRA/DO approximation of (1) is defined by the Dirac–Frenkel principle [23, 31]

$$\frac{d\tilde{X}}{dt} = P_{T_{\tilde{X}}\mathcal{M}_r}F(\tilde{X}, t) = \arg \min_{V \in T_{\tilde{X}}\mathcal{M}_r} \|V - F(\tilde{X}, t)\|. \quad (4)$$

The representation $\delta U Z^\top + U \delta Z^\top$ of a tangent vector is not unique; the DO gauge $\dot{U} \in \mathcal{U}_{m,r} = \{\delta U : U^\top \delta U = 0\}$ [68] removes the redundancy and yields the DO equations [59, 68]

$$\dot{U} = P_U^\perp F(UZ^\top, t) Z (Z^\top Z)^{-1}, \quad \dot{Z} = F(UZ^\top, t)^\top U. \quad (5)$$

Justification. Writing $\dot{\tilde{X}} = \dot{U}Z^\top + U\dot{Z}^\top$ with $U^\top \dot{U} = 0$, the two terms are orthogonal; minimizing $\|F - \dot{U}Z^\top - U\dot{Z}^\top\|^2 = \|P_U F - U\dot{Z}^\top\|^2 + \|P_U^\perp F - \dot{U}Z^\top\|^2$ separately gives $\dot{Z} = F^\top U$ and the normal equations $\dot{U}Z^\top Z = P_U^\perp FZ$. Substituting back, $\dot{U}Z^\top + U\dot{Z}^\top = P_U^\perp F P_Z + P_U F = P_{T_{\tilde{X}}\mathcal{M}_r}F$, which confirms (3). In the KL gauge the DO mode velocity of mode j is $\dot{u}_j = P_U^\perp F z_j / \sigma_j^2$ and the coefficient velocity is $\dot{z}_j = F^\top u_j$; we use these two vectors to define the staleness scores in Section 4.

The DLRA is exact when $F(\tilde{X}, t) \in T_{\tilde{X}}\mathcal{M}_r$; otherwise the neglected *normal component* $(I - P_{T_{\tilde{X}}\mathcal{M}_r})F = P_U^\perp F P_Z^\perp$ is the source of the approximation error. Koch and Lubich [48] showed that if $\|(I - P_{T_{\tilde{X}}\mathcal{M}_r})F\| \leq \varepsilon$ along the trajectory, F is Lipschitz with constant ℓ and the singular values stay bounded away from zero, then $\|\tilde{X}(t) - X(t)\| \leq C(e^{\ell t} - 1)\varepsilon/\ell$; Kieri et al. [47] and Ceruti and Lubich [10] obtained bounds of the same form for the projector-splitting and BUG integrators that are independent of the smallest singular value. Section 5 extends these bounds to the selective dynamics.

3.3 Projection methods, retractions and DORK schemes

Rather than integrating (5) directly, projection methods [13, 46] integrate the full-rank increment and return to the manifold. Following Charous and Lermusiaux [14], let

$$\Phi = \frac{1}{\Delta t} \int_{t_i}^{t_{i+1}} F(X(t), t) dt + \mathcal{O}(\Delta t^k) \quad (6)$$

denote the exactly or approximately (order k) integrated operator over one step, so that $X_{i+1} = X_i + \Delta t \Phi$ represents any chosen k th-order integration scheme, and let $\chi = \tilde{X}_i + \Delta t \Phi$ be the *before retraction point*. The best rank- r approximation of χ is $P_{\mathcal{M}_r}(\chi)$, the truncated SVD [27, 58], but computing it every step is expensive; a *retraction* $\mathcal{R}_{\tilde{X}}(\Delta t \Phi) = \tilde{X} + \Delta t P_{T_{\tilde{X}}\mathcal{M}_r} \Phi + \mathcal{O}(\Delta t^2)$ [1, 2] replaces it, and the scheme becomes

$$\tilde{X}_{i+1} = \mathcal{R}_{\tilde{X}_i}(\Delta t \Phi). \quad (7)$$

Charous and Lermusiaux [13, 14] solve the projection problem

$$\min_{U_{i+1}, Z_{i+1}} \left\| \chi - U_{i+1} Z_{i+1}^\top \right\| \quad \text{s.t.} \quad U_{i+1} = (U_i + \Delta t \dot{U}) G, \quad \dot{U} \in \mathcal{U}_{m,r}, \quad Z_{i+1} = \chi^\top U_{i+1}, \quad (8)$$

where G re-orthonormalizes, by expanding $\Delta t \dot{U} = \sum_{j \geq 1} \Delta t^j \dot{U}_j$ in powers of Δt ; grouping orders gives the *optimal perturbative retractions* [14, Eq. (2.14)], whose first-order term is

$$\dot{U}_1 = P_{U_i}^\perp \Phi Z_i (Z_i^\top Z_i)^{-1}, \quad (9)$$

i.e., the DO mode equation (5) with F replaced by Φ , and whose coefficient update $Z_{i+1} = \chi^\top U_{i+1}$ is the optimal least-squares fit given the new basis, the same update as in POD and other reduced-basis methods [14]. A q th-order retraction has projection–retraction error $\varepsilon_{pr} = \mathcal{R}_{\tilde{X}_i}(\Delta t \Phi) - P_{\mathcal{M}_r}(\chi) = \mathcal{O}(\Delta t^{q+1})$. Key properties proved in Charous and Lermusiaux [14] and used below are:

- *Stability* [14, Lemma 2.2.1]: $\left\| \tilde{X}_{i+1} \right\| \leq \left\| \tilde{X}_i + \Delta t \Phi \right\|$, because $\|Z_{i+1}\| = \|\chi^\top U_{i+1}\| \leq \|\chi\|$.
- *Gradient-descent retractions* [14, Alg. 2.1–2.2]: iterating $\tilde{X}_{i+1} \leftarrow \mathcal{R}_{\tilde{X}_{i+1}}(\Delta t \Phi - (\tilde{X}_{i+1} - \tilde{X}_i))$ converges superlinearly to a rank- r target and linearly to the best rank- r approximation of a higher-rank target.
- *Robust retractions* [14, Eqs. (2.22)–(2.24)]: $U_{i+1} = \text{orth}(U_i(Z_i^\top Z_i) + \Delta t P_{U_i}^\perp \Phi Z_i)$, $Z_{i+1} = \chi^\top U_{i+1}$, avoid inverting an ill-conditioned $Z_i^\top Z_i$ at the price of mode continuity, or, alternatively, the pseudo-inverse preserves mode continuity [5].
- *Rank adaptation* [14, Alg. 2.3]: the rank is increased when the angle $\theta = \cos^{-1}(\|P_{T_{\tilde{X}} \mathcal{M}_r} \Phi\| / \|\Phi\|)$ exceeds a threshold, by appending leading left singular vectors Q of $P_U^\perp \Phi$ (randomized SVD, 41) with zero coefficients that receive the kick $\hat{Z} \leftarrow \hat{Z} + \Delta t \Phi^\top \hat{U}$; it is decreased by discarding the smallest eigenpairs of $Z^\top Z$ whose cumulative energy falls below σ_* [11, 69].
- *DORK schemes* [14, Sec. 2.5]: expanding the integrated operator as $\Delta t \Phi = \sum_{j \geq 1} \Delta t^j \Phi_j$, with partial sums $\Phi^{(k)} = \sum_{j \leq k} \Delta t^{j-1} \Phi_j$ forming a k th-order Runge–Kutta scheme, and substituting into (8) yields the stable, optimal so-DORK schemes; applying the gradient-descent retraction successively to $\Phi^{(1)}, \dots, \Phi^{(k)}$ yields the gd-DORK schemes, which are consistent and k th-order accurate [14, Lemma 2.5.2].

3.4 Error taxonomy

Table 2 recalls the error components of Charous and Lermusiaux [14] and adds the two new components introduced by selective updates. The normal closure error $\varepsilon_{\mathcal{N}}$ and the dynamical model closure error ε_D are modeling errors that any rank- r approximation incurs; ε_{DF} is the instantaneous Dirac–Frenkel defect; ε_l and ε_{pr} are discretization errors of the retraction; ε_S and ε_Z measure the effect of freezing bases and coefficients.

4 Selective-update dynamically orthogonal dynamics

4.1 Active, frozen and dormant modes

Let $[r] = \{1, \dots, r\}$ index the columns of U and Z . At any instant we partition the modes into three *tiers* by means of two index sets:

Table 2: Error components. The first five follow Charous and Lermusiaux [14, Table 2.1]; the last two are introduced in this paper (Section 4). X is the full-rank state, $\tilde{X}$ the rank- r approximation, $\chi = \tilde{X}_i + \Delta t \Phi$.

Symbol	Name	Definition
$\varepsilon_{\mathcal{N}}$	normal closure error	$X - P_{\mathcal{M}_r}(X)$
ε_D	dynamical model closure error	$F(X) - F(\tilde{X})$
ε_{DF}	Dirac–Frenkel defect	$F(\tilde{X}) - P_{T_{\tilde{X}}\mathcal{M}_r}F(\tilde{X}) = P_U^\perp F P_Z^\perp$
ε_l	local retraction error	$\tilde{X}_i + \Delta t \Phi - \mathcal{R}_{\tilde{X}_i}(\Delta t \Phi)$
ε_{pr}	projection–retraction error	$\mathcal{R}_{\tilde{X}_i}(\Delta t \Phi) - P_{\mathcal{M}_r}(\chi)$
ε_S	basis-staleness defect	$P_U^\perp F (P_Z - P_{Z_A})$
ε_Z	coefficient-staleness defect	$U_{\mathcal{D}} U_{\mathcal{D}}^\top F$

- $\mathcal{A} \subseteq [r]$, the *active* set: modes whose basis vectors u_j are advanced; its complement $\mathcal{F} = [r] \setminus \mathcal{A}$ is the *frozen* set, whose basis vectors are held fixed; $a = |\mathcal{A}|$, $f = |\mathcal{F}| = r - a$.
- $\mathcal{B} \subseteq [r]$, the set of modes whose coefficient columns z_j are advanced; its complement $\mathcal{D} = [r] \setminus \mathcal{B}$ is the *dormant* set, whose coefficients are held fixed; $b = |\mathcal{B}|$.

A mode $j \in \mathcal{A} \cap \mathcal{B}$ is fully *dynamic* (tier 2, the ordinary DO mode); a mode $j \in \mathcal{F} \cap \mathcal{B}$ is a *Galerkin* mode (tier 1: fixed basis vector, evolving coefficient, as in a POD–Galerkin model); a mode $j \in \mathcal{F} \cap \mathcal{D}$ is *dormant* (tier 0: its rank-one contribution $u_j z_j^\top$ is carried along unchanged). The combination $j \in \mathcal{A} \cap \mathcal{D}$ (moving basis vector with frozen coefficient) is admissible and is in fact how a deterministic mean field enters the matrix formulation (Remark 4.3), although the scheduler of Section 6 rarely selects it for stochastic modes because coefficient updates are cheap. We write $U = [U_{\mathcal{A}} \ U_{\mathcal{F}}]$ and $Z = [Z_{\mathcal{A}} \ Z_{\mathcal{F}}]$ up to a column permutation, and similarly for $\mathcal{B}, \mathcal{D}$. The sets $\mathcal{A}, \mathcal{B}$ and the rank r are piecewise constant in time; they change only at the scheduling instants of Section 6.

4.2 Constrained variational principle and selective DO equations

Freezing the basis vectors $U_{\mathcal{F}}$ restricts the approximation to the set of rank- r matrices whose column space contains the fixed subspace $\mathcal{V}_{\mathcal{F}} = \text{range}(U_{\mathcal{F}})$,

$$\mathcal{M}_r^{\mathcal{F}} = \{X \in \mathcal{M}_r : \mathcal{V}_{\mathcal{F}} \subseteq \text{range } X\} = \{U_{\mathcal{F}} Z_{\mathcal{F}}^\top + U_{\mathcal{A}} Z_{\mathcal{A}}^\top : U_{\mathcal{A}} \in \mathcal{V}_{m,a}, U_{\mathcal{F}}^\top U_{\mathcal{A}} = 0, Z \in \mathbb{R}_*^{n \times r}\}. \quad (10)$$

This is a smooth embedded submanifold of $\mathcal{M}_r$ of dimension $(m-r)a + nr$ (it is the image of the product of the Stiefel manifold of $\mathcal{V}_{\mathcal{F}}^\perp$ and $\mathbb{R}_*^{n \times r}$ under a submersion with $\mathcal{O}(a)$ fibers; the count follows from $(m-f)a - a^2 + nr$ and reduces to $(m+n-r)r$ when $f=0$). Its tangent space at $\tilde{X} = UZ^\top$ in the DO gauge is

$$T_{\tilde{X}} \mathcal{M}_r^{\mathcal{F}} = \{\delta U_{\mathcal{A}} Z_{\mathcal{A}}^\top + U \delta Z^\top : U^\top \delta U_{\mathcal{A}} = 0\}, \quad (11)$$

and freezing the dormant coefficients further restricts admissible velocities to the linear subspace

$$\mathcal{S}_{\tilde{X}}^{\mathcal{A},\mathcal{B}} = \{\delta U_{\mathcal{A}} Z_{\mathcal{A}}^\top + U_{\mathcal{B}} \delta Z_{\mathcal{B}}^\top : U^\top \delta U_{\mathcal{A}} = 0, \delta Z_{\mathcal{B}} \in \mathbb{R}^{n \times b}\} \subseteq T_{\tilde{X}} \mathcal{M}_r^{\mathcal{F}} \subseteq T_{\tilde{X}} \mathcal{M}_r. \quad (12)$$

The *selective DO approximation* is defined by the Dirac–Frenkel principle on $\mathcal{S}_{\tilde{X}}^{\mathcal{A},\mathcal{B}}$:

$$\frac{d\tilde{X}}{dt} = P_{\tilde{X}}^{\mathcal{A},\mathcal{B}} F(\tilde{X}, t), \quad P_{\tilde{X}}^{\mathcal{A},\mathcal{B}} F = \arg \min_{V \in \mathcal{S}_{\tilde{X}}^{\mathcal{A},\mathcal{B}}} \|V - F\|. \quad (13)$$

Proposition 4.1 (Selective projector and selective DO equations). *Let $\tilde{X} = UZ^\top$ with $U \in \mathcal{V}_{m,r}$, $Z \in \mathbb{R}_*^{n \times r}$, and let $P_{Z_A} = Z_A(Z_A^\top Z_A)^{-1}Z_A^\top$. The orthogonal projector onto $\mathcal{S}_{\tilde{X}}^{A,B}$ is*

$$P_{\tilde{X}}^{A,B}F = U_B U_B^\top F + P_U^\perp F P_{Z_A}, \quad (14)$$

and the minimizer in (13) is realized by the selective DO equations

$$\dot{U}_A = P_U^\perp F Z_A (Z_A^\top Z_A)^{-1}, \quad \dot{U}_F = 0, \quad \dot{Z}_B = F^\top U_B, \quad \dot{Z}_D = 0. \quad (15)$$

Proof. The two summands of a generic element of $\mathcal{S}_{\tilde{X}}^{A,B}$ are orthogonal, $\langle \delta U_A Z_A^\top, U_B \delta Z_B^\top \rangle = \text{tr}(Z_A \delta U_A^\top U_B \delta Z_B^\top) = 0$, because $\delta U_A^\top U = 0$. Hence

$$\left\| F - \delta U_A Z_A^\top - U_B \delta Z_B^\top \right\|^2 = \left\| P_U F - U_B \delta Z_B^\top \right\|^2 + \left\| P_U^\perp F - \delta U_A Z_A^\top \right\|^2,$$

and the two terms can be minimized separately. Since the columns of U_B are orthonormal, the first is minimized by $\delta Z_B^\top = U_B^\top F$. The second is a linear least-squares problem in δU_A whose normal equations read $\delta U_A Z_A^\top Z_A = P_U^\perp F Z_A$; its solution $\delta U_A = P_U^\perp F Z_A (Z_A^\top Z_A)^{-1}$ satisfies the gauge constraint $U^\top \delta U_A = 0$ automatically. Substituting gives (14). $\square$

Remark 4.2 (Active versus full covariance). The selective mode velocity uses the *active* covariance $Z_A^\top Z_A$, not the $\mathcal{A}$ -block of $(Z^\top Z)^{-1}$. By the block inverse formula, $[P_U^\perp F Z (Z^\top Z)^{-1}]_A = P_U^\perp F (Z_A - P_{Z_F} Z_A) (Z_A^\top P_{Z_F}^\perp Z_A)^{-1}$, which coincides with (15) if and only if $Z_F^\top Z_A = 0$, i.e., in the KL gauge. In a general gauge the selective velocity is the least-squares optimal correction *given that the frozen modes do not move*: it partially compensates for the frozen rotation through the correlation between active and frozen coefficients. Both are consistent with the Dirac–Frenkel principle on their respective constraint sets; the scheduler always rotates to the KL gauge at decision instants (Section 6), where the two coincide, and the gauge then drifts slowly within a scheduling interval.

Remark 4.3 (The stochastic mean). In the DO expansion $u = \bar{u} + \sum_j Y_j u_j$ with $\mathbb{E}[Y_j] = 0$, the mean $\bar{u}$ is a deterministic field that is always advanced. In the matrix formulation it is the mode $u_0 = \bar{u} / \|\bar{u}\|$ with the *fixed* coefficient column $z_0 = \|\bar{u}\| \mathbf{1} / \sqrt{n}$, orthogonal to the centered columns z_j ($\mathbf{1}^\top z_j = \sqrt{n} \mathbb{E}[Y_j] = 0$); it is thus a permanently active-basis, frozen-coefficient mode, $0 \in \mathcal{A} \cap \mathcal{D}$, and (15) gives $\dot{u}_0 = P_U^\perp \mathbb{E}[F] \sqrt{n} / \|\bar{u}\|$, the DO mean equation up to the component in range U that the coefficient equations absorb. Alternatively the mean can be evolved separately as in Sapsis and Lermusiaux [68], Ueckermann et al. [77]; the analysis below is unaffected. We therefore state the results for $\tilde{X} = UZ^\top$ and refer to the mean only where the cost model requires it.

4.3 Exact decomposition of the selective defect

Theorem 4.4 (Defect decomposition). *Let $\tilde{X} = UZ^\top$ and define $\tilde{Z}_F = P_{Z_A}^\perp Z_F$. The defect of the selective projection decomposes orthogonally as*

$$F - P_{\tilde{X}}^{A,B}F = \underbrace{P_U^\perp F P_Z^\perp}_{\varepsilon_{DF}} + \underbrace{P_U^\perp F P_{\tilde{Z}_F}}_{\varepsilon_S} + \underbrace{U_D U_D^\top F}_{\varepsilon_Z}, \quad (16)$$

$$\left\| F - P_{\tilde{X}}^{A,B}F \right\|^2 = \|\varepsilon_{DF}\|^2 + \|\varepsilon_S\|^2 + \|\varepsilon_Z\|^2.$$

In the KL gauge ($Z^\top Z = \text{diag}(\sigma_j^2)$), with the DO velocities $\dot{u}_j = P_U^\perp F z_j / \sigma_j^2$ and $\dot{z}_j = F^\top u_j$ of (5),

$$\begin{aligned} \|\varepsilon_S\|^2 &= \sum_{j \in \mathcal{F}} s_j^2, & s_j &:= \frac{\|P_U^\perp F z_j\|}{\sigma_j} = \|P_U^\perp F v_j\| = \sigma_j \|\dot{u}_j\|; \\ \|\varepsilon_Z\|^2 &= \sum_{j \in \mathcal{D}} c_j^2, & c_j &:= \|F^\top u_j\| = \|\dot{z}_j\|. \end{aligned} \quad (17)$$

Proof. By (14), $F - P_{\tilde{X}}^{\mathcal{A}, \mathcal{B}} F = (I - U_{\mathcal{B}} U_{\mathcal{B}}^\top) F - P_U^\perp F P_{Z_{\mathcal{A}}} = P_U^\perp F (I - P_{Z_{\mathcal{A}}}) + U_{\mathcal{D}} U_{\mathcal{D}}^\top F$, using $I - U_{\mathcal{B}} U_{\mathcal{B}}^\top = P_U^\perp + U_{\mathcal{D}} U_{\mathcal{D}}^\top$. Since $\text{range } Z_{\mathcal{A}} \subseteq \text{range } Z$ and $\text{range } Z = \text{range } Z_{\mathcal{A}} \oplus \text{range } \tilde{Z}_{\mathcal{F}}$ (Gram–Schmidt of $Z_{\mathcal{F}}$ against $Z_{\mathcal{A}}$; $\text{rank } \tilde{Z}_{\mathcal{F}} = f$ because Z has full column rank), the projectors satisfy $P_Z = P_{Z_{\mathcal{A}}} + P_{\tilde{Z}_{\mathcal{F}}}$ and $I - P_{Z_{\mathcal{A}}} = P_Z^\perp + P_{\tilde{Z}_{\mathcal{F}}}^\perp$, which gives the three-term identity. Orthogonality: $\langle P_U^\perp F P_Z^\perp, P_U^\perp F P_{\tilde{Z}_{\mathcal{F}}} \rangle = \text{tr}(P_Z^\perp F^\top P_U^\perp F P_{\tilde{Z}_{\mathcal{F}}}) = \text{tr}(F^\top P_U^\perp F P_{\tilde{Z}_{\mathcal{F}}} P_Z^\perp) = 0$ since $P_{\tilde{Z}_{\mathcal{F}}} P_Z^\perp = 0$; and the third term is orthogonal to the first two because its column space lies in $\text{range } U_{\mathcal{D}} \subseteq \text{range } U$, whereas theirs lie in $\text{range } U^\perp$. In the KL gauge $\tilde{Z}_{\mathcal{F}} = Z_{\mathcal{F}}$ and $Q_{\mathcal{F}} = Z_{\mathcal{F}} \Lambda_{\mathcal{F}}^{-1/2}$ has orthonormal columns, so $\|P_U^\perp F P_{Z_{\mathcal{F}}}\|^2 = \|P_U^\perp F Q_{\mathcal{F}}\|^2 = \sum_{j \in \mathcal{F}} \|P_U^\perp F z_j\|^2 / \sigma_j^2$, and $\|U_{\mathcal{D}} U_{\mathcal{D}}^\top F\|^2 = \sum_{j \in \mathcal{D}} \|u_j^\top F\|^2$. The identities $s_j = \sigma_j \|\dot{u}_j\|$ and $c_j = \|\dot{z}_j\|$ follow from (5). $\square$

Here $v_j = z_j / \sigma_j$ is the j th right singular vector of $\tilde{X}$, so that s_j is the normal component of the dynamics along the j th stochastic direction; this form is the one to evaluate numerically, since it stays bounded as $\sigma_j \rightarrow 0$ whereas $\dot{u}_j$ does not.

Theorem 4.4 has a transparent reading. The first term is the classical Dirac–Frenkel defect that any rank- r approximation incurs [14, 48]. The price of freezing the basis vector of mode j is its *amplitude times its rotation rate*, $s_j = \sigma_j \|\dot{u}_j\|$: a strong mode that rotates slowly and a weak mode that rotates quickly are equally cheap to freeze. The price of freezing coefficient j is its *own rate of change* $c_j = \|\dot{z}_j\|$, the norm of the projection of the dynamics onto the fixed direction u_j . Both scores are by-products of quantities the full DO step computes anyway.

Corollary 4.5 (Selective angle criterion). *Let $\theta = \cos^{-1}(\|P_{T_{\tilde{X}} \mathcal{M}_r} F\| / \|F\|)$ be the angle of Charous and Lermusiaux [14] between F and the tangent space, and $\theta^{\mathcal{A}, \mathcal{B}} = \cos^{-1}(\|P_{\tilde{X}}^{\mathcal{A}, \mathcal{B}} F\| / \|F\|)$ its selective counterpart. Then*

$$\sin^2 \theta^{\mathcal{A}, \mathcal{B}} = \sin^2 \theta + \frac{\sum_{j \in \mathcal{F}} s_j^2 + \sum_{j \in \mathcal{D}} c_j^2}{\|F\|^2}. \quad (18)$$

Proof. $\|F\|^2 = \|P_{\tilde{X}}^{\mathcal{A}, \mathcal{B}} F\|^2 + \|F - P_{\tilde{X}}^{\mathcal{A}, \mathcal{B}} F\|^2$ and $\|F\|^2 \sin^2 \theta = \|\varepsilon_{DF}\|^2$; insert (16)–(17). $\square$

Lemma 4.6 (Optimal active sets by sorting). *Let $s, c \in \mathbb{R}_{\geq 0}^r$ be the scores of (17) and $0 \leq a, b \leq r$. Among all $\mathcal{A}$ with $|\mathcal{A}| = a$ and $\mathcal{B}$ with $|\mathcal{B}| = b$, the staleness defect $\|\varepsilon_S\|^2 + \|\varepsilon_Z\|^2 = \sum_{j \notin \mathcal{A}} s_j^2 + \sum_{j \notin \mathcal{B}} c_j^2$ is minimized by taking $\mathcal{A}$ to be the indices of the a largest s_j and $\mathcal{B}$ those of the b largest c_j . If in addition the nesting $\mathcal{A} \subseteq \mathcal{B}$ is imposed, the minimizer solves a three-tier assignment (transportation) problem with tier weights $s_j^2 + c_j^2$, c_j^2 and 0 and tier capacities a , $b - a$ and $r - b$, whose linear-programming relaxation is integral; for fixed $\mathcal{A}$ its optimal $\mathcal{B}$ is $\mathcal{A}$ together with the $b - a$ largest c_j over $j \notin \mathcal{A}$.*

Proof. The objective is separable, so the two sums are minimized independently; a sum of $r - a$ nonnegative terms taken from a fixed multiset is smallest when it omits the a largest terms (exchange

argument). Under nesting, write $\sum_{j \in \mathcal{A}} (s_j^2 + c_j^2) + \sum_{j \in \mathcal{B} \setminus \mathcal{A}} c_j^2$ for the quantity to be maximized; assigning each j to exactly one of three tiers with prescribed cardinalities is a transportation problem, whose constraint matrix is totally unimodular [28, Ch. 7], and for fixed $\mathcal{A}$ the $\mathcal{B}$ -part is again a top- $(b - a)$ selection. $\square$

Because the coefficient update is by far the cheapest operation of a step (Section 6.1), the scheduler almost always selects $b = r$, in which case the nesting constraint is automatically satisfied and the sorting rule is exact.

4.4 Discrete step: selective retractions and the deflation theorem

We now discretize (13) in the projection–retraction framework of Section 3.3. Given $\tilde{X}_i = U_i Z_i^\top$, the integrated operator Φ and $\chi = \tilde{X}_i + \Delta t \Phi$, the *selective retraction problem* is

$$\begin{aligned} \min_{\dot{U}_{\mathcal{A}}, Z_{i+1, \mathcal{B}}} \quad & \left\| \chi - U_{i+1} Z_{i+1}^\top \right\| \\ \text{s.t.} \quad & U_{i+1} = [U_{i, \mathcal{F}} \quad (U_{i, \mathcal{A}} + \Delta t \dot{U}_{\mathcal{A}}) G], \quad U_i^\top \dot{U}_{\mathcal{A}} = 0, \quad G \in \mathbb{R}_*^{a \times a}, \\ & Z_{i+1, \mathcal{B}} = \chi^\top U_{i+1, \mathcal{B}}, \quad Z_{i+1, \mathcal{D}} = Z_{i, \mathcal{D}}, \end{aligned} \quad (19)$$

where G re-orthonormalizes the active block. As in (8), the coefficient constraint is the exact minimizer over $Z_{i+1, \mathcal{B}}$ for any admissible U_{i+1} (the columns of U_{i+1} are orthonormal and $U_{i+1, \mathcal{B}}^\top U_{i, \mathcal{D}} = 0$), so the problem is an optimization over the active basis only.

Theorem 4.7 (Deflation). *Assume $\mathcal{D} \subseteq \mathcal{F}$ (every mode whose basis moves has its coefficient updated; the deterministic mean of Remark 4.3 is evolved separately). Define the deflated before retraction point and increment*

$$\chi' = P_{U_{i, \mathcal{F}}}^\perp \chi = U_{i, \mathcal{A}} Z_{i, \mathcal{A}}^\top + \Delta t \Phi', \quad \Phi' = P_{U_{i, \mathcal{F}}}^\perp \Phi. \quad (20)$$

(i) *For every admissible U_{i+1} in (19),*

$$\left\| \chi - U_{i+1} Z_{i+1}^\top \right\|^2 = \left\| (I - U_{i+1, \mathcal{A}} U_{i+1, \mathcal{A}}^\top) \chi' \right\|^2 + \left\| U_{i, \mathcal{D}}^\top \chi - Z_{i, \mathcal{D}}^\top \right\|^2, \quad (21)$$

and the second term does not depend on $\dot{U}_{\mathcal{A}}$. Hence (19) is the optimal perturbative retraction problem (8) of Charous and Lermusiaux [14] for the rank- a matrix $\tilde{X}'_i = U_{i, \mathcal{A}} Z_{i, \mathcal{A}}^\top$ in the direction Φ' .

(ii) *The perturbative corrections $\dot{U}_1, \dot{U}_2, \dots$ of Charous and Lermusiaux [14, Eqs. (2.14), (2.28)] evaluated for $(\tilde{X}'_i, \Phi')$ are automatically orthogonal to $U_{i, \mathcal{F}}$; in particular the first-order selective correction is*

$$\dot{U}_{1, \mathcal{A}} = P_{U_i}^\perp \Phi Z_{i, \mathcal{A}} (Z_{i, \mathcal{A}}^\top Z_{i, \mathcal{A}})^{-1}, \quad (22)$$

the discrete counterpart of (15).

(iii) *Consequently, for any retraction $\mathcal{R}$ of the family of Charous and Lermusiaux [13, 14] (optimal perturbative of any order, gradient-descent, robust, or their so-/gd-DORK stage versions), the selective retraction*

$$\begin{aligned} U_{i+1, \mathcal{A}} Z_{i+1}^\top &= \mathcal{R}_{\tilde{X}'_i}(\Delta t \Phi'), & U_{i+1} &= [U_{i, \mathcal{F}} \quad U_{i+1, \mathcal{A}}], \\ Z_{i+1, \mathcal{B}} &= \chi^\top U_{i+1, \mathcal{B}}, & Z_{i+1, \mathcal{D}} &= Z_{i, \mathcal{D}}, \end{aligned} \quad (23)$$

has the same projection–retraction order as $\mathcal{R}$ with respect to the constrained best approximation $\arg \min \{ \|\chi - Y\| : Y \in \mathcal{M}_r^{\mathcal{F}_i}, Y^\top U_{i, \mathcal{D}} = Z_{i, \mathcal{D}} \}$, and $Z'_{i+1} = Z_{i+1, \mathcal{A}}$.

Proof. (i) U_{i+1} has orthonormal columns with $U_{i+1,\mathcal{A}} \perp U_{i,\mathcal{F}}$, so $P_{U_{i+1}}^\perp \chi = \chi - U_{i,\mathcal{F}} U_{i,\mathcal{F}}^\top \chi - U_{i+1,\mathcal{A}} U_{i+1,\mathcal{A}}^\top \chi = (I - U_{i+1,\mathcal{A}} U_{i+1,\mathcal{A}}^\top) \chi'$, using $U_{i+1,\mathcal{A}}^\top \chi = U_{i+1,\mathcal{A}}^\top \chi'$. Substituting the coefficient constraint and $U_{i+1,\mathcal{D}} = U_{i,\mathcal{D}}$ (dormant modes are frozen), $\chi - U_{i+1} Z_{i+1}^\top = P_{U_{i+1}}^\perp \chi + U_{i,\mathcal{D}} (U_{i,\mathcal{D}}^\top \chi - Z_{i,\mathcal{D}}^\top)$. The two summands are orthogonal since the first lies in range $U_{i+1}^\perp$ and the second in range U_{i+1} , which gives (21). Minimizing $\|(I - U_{i+1,\mathcal{A}} U_{i+1,\mathcal{A}}^\top) \chi'\|$ over $U_{i+1,\mathcal{A}} = (U_{i,\mathcal{A}} + \Delta t \dot{U}_{\mathcal{A}}) G$ with $\dot{U}_{\mathcal{A}} \perp U_i$ is exactly problem (8) for $(\tilde{X}'_i, \Phi')$ with the extra constraint $\dot{U}_{\mathcal{A}} \perp U_{i,\mathcal{F}}$, and by (ii) this constraint is inactive. (ii) Every correction in Charous and Lermusiaux [14, Eqs. (2.14), (2.28)] is of the form $P_{U_{i,\mathcal{A}}}^\perp [\cdot] (Z_{i,\mathcal{A}}^\top Z_{i,\mathcal{A}})^{-1}$, where $[\cdot]$ is a sum of terms each of which is either $\Phi'(\dots)$, $\Phi' \Phi'^\top(\dots)$, or a lower-order correction $\dot{U}_k(\dots)$ right-multiplied by an $a \times a$ matrix. Since $\Phi' = P_{U_{i,\mathcal{F}}}^\perp \Phi$ and, inductively, $\dot{U}_k \perp U_{i,\mathcal{F}}$, each term is orthogonal to $U_{i,\mathcal{F}}$, and $P_{U_{i,\mathcal{A}}}^\perp$ preserves this property. For the first order, $P_{U_{i,\mathcal{A}}}^\perp P_{U_{i,\mathcal{F}}}^\perp = P_{U_i}^\perp$ gives (22). (iii) The constrained best approximation is $U_{i,\mathcal{F}} Z_{i,\mathcal{F}}^{*\top} + P_{\mathcal{M}_a}(\chi')$ with $Z_{i,\mathcal{F}}^*$ determined by the coefficient constraints, because (21) is minimized when $U_{i+1,\mathcal{A}}$ spans the leading left singular subspace of χ' [27]; the projection–retraction error of (23) is therefore that of $\mathcal{R}$ on the rank- a problem, and $Z'_{i+1} = \chi'^\top U_{i+1,\mathcal{A}} = \chi^\top U_{i+1,\mathcal{A}}$. $\square$

Theorem 4.7 is the practical heart of the method: a selective step is implemented by calling the existing (rank-adaptive) DORK step on the active block with the deflated increment $P_{U_{\mathcal{F}}}^\perp \Phi$, then applying the POD-like coefficient update (23) to the Galerkin modes and leaving the dormant ones untouched. The *robust* first-order selective retraction is

$$U_{i+1,\mathcal{A}} = \text{orth}_{\perp U_{i,\mathcal{F}}} (U_{i,\mathcal{A}} (Z_{i,\mathcal{A}}^\top Z_{i,\mathcal{A}}) + \Delta t P_{U_i}^\perp \Phi Z_{i,\mathcal{A}}), \quad Z_{i+1,\mathcal{B}} = \chi^\top U_{i+1,\mathcal{B}}, \quad (24)$$

where $\text{orth}_{\perp U_{\mathcal{F}}}$ orthonormalizes against the frozen block first (Gram–Schmidt with re-orthogonalization, 19, or the fast re-orthogonalization of 77) and then internally; it never inverts $Z_{i,\mathcal{A}}^\top Z_{i,\mathcal{A}}$ and is therefore the retraction to use immediately after rank augmentation, when newly added active modes have vanishing singular values [14, Sec. 2.4]. For a Galerkin mode $j \in \mathcal{F} \cap \mathcal{B}$, (23) reduces to the explicit update

$$z_{i+1,j} = \chi^\top u_j = z_{i,j} + \Delta t \Phi^\top u_j, \quad (25)$$

which is the Galerkin projection of the k th-order integrated dynamics onto the fixed direction u_j , exactly as in a POD–Galerkin reduced-order model [44, 50] advanced with the same Runge–Kutta scheme.

Lemma 4.8 (Stability of the selective step). *Let $\tilde{X}_{i+1}$ be produced by (23) with any retraction of Charous and Lermusiaux [14], and let $\chi = \tilde{X}_i + \Delta t \Phi$.*

(a) *If $\mathcal{B} = [r]$ (no dormant coefficients), then $\|\tilde{X}_{i+1}\| \leq \|\chi\|$.*

(b) *In general, with $c_{\mathcal{D}} = (\sum_{j \in \mathcal{D}} c_j^2)^{1/2} = \|\Phi^\top U_{i,\mathcal{D}}\|$ and $\mathcal{D} \subseteq \mathcal{F}$,*

$$\|\tilde{X}_{i+1}\|^2 = \|U_{i+1,\mathcal{B}}^\top \chi\|^2 + \|Z_{i,\mathcal{D}}\|^2 \leq \|\chi\|^2 + 2\Delta t \|Z_{i,\mathcal{D}}\| c_{\mathcal{D}}. \quad (26)$$

Proof. (a) $\|\tilde{X}_{i+1}\| = \|Z_{i+1}\| = \|\chi^\top U_{i+1}\| \leq \|\chi\|$ because U_{i+1} has orthonormal columns; this is Charous and Lermusiaux [14, Lemma 2.2.1] applied to the full basis U_{i+1} , which (23) keeps orthonormal. (b) $\|\tilde{X}_{i+1}\|^2 = \|Z_{i+1,\mathcal{B}}\|^2 + \|Z_{i,\mathcal{D}}\|^2$ and $\|\chi\|^2 = \|U_{i+1,\mathcal{B}}^\top \chi\|^2 + \|U_{i,\mathcal{D}}^\top \chi\|^2 + \|P_{U_{i+1}}^\perp \chi\|^2$. Since $U_{i,\mathcal{D}}^\top \chi = Z_{i,\mathcal{D}}^\top + \Delta t U_{i,\mathcal{D}}^\top \Phi$, $\|Z_{i,\mathcal{D}}\|^2 - \|U_{i,\mathcal{D}}^\top \chi\|^2 = -2\Delta t \langle Z_{i,\mathcal{D}}, \Phi^\top U_{i,\mathcal{D}} \rangle - \Delta t^2 \|\Phi^\top U_{i,\mathcal{D}}\|^2 \leq 2\Delta t \|Z_{i,\mathcal{D}}\| c_{\mathcal{D}}$ by Cauchy–Schwarz. $\square$

Part (a) states that the stability property of the so-DORK schemes [14, Lemma 2.5.1], low-rank stability whenever the underlying Runge–Kutta scheme is stable, is inherited by any selection of active bases, as long as all coefficients are updated. Part (b) quantifies the only way dormancy can violate it: a dormant coefficient retains energy that the dynamics would have removed, by at most $2\Delta t \|Z_{i,\mathcal{D}}\| c_{\mathcal{D}}$ per step, which the scheduler controls through the threshold on c_j (Section 6). Dormancy is therefore reserved for modes whose coefficient velocity is negligible.

4.5 Selective so-DORK and gd-DORK schemes

The DORK schemes expand the integrated operator as $\Delta t \Phi = \sum_{j \geq 1} \Delta t^j \Phi_j$ [14, Eq. (2.27)], where the Runge–Kutta stages are evaluated on the low-rank state, that is, on the *full* $\tilde{X} = UZ^\top$ including its frozen and dormant components. The deflation acts on the increments only: substituting $\Phi'_j = P_{U_{i,\mathcal{F}}}^\perp \Phi_j$, $U_{i,\mathcal{A}}$ and $Z_{i,\mathcal{A}}$ into the so-DORK corrections [14, Eq. (2.28)] yields the *selective so-DORK* schemes; applying the deflated gradient-descent retraction successively to $\Phi'^{(1)}, \dots, \Phi'^{(k)}$ (Algorithm 1 below) yields the *selective gd-DORK* schemes. Two properties follow directly from Theorem 4.7: the selective so-DORK schemes satisfy Lemma 4.8, and the selective gd-DORK schemes are consistent and k th-order accurate with respect to the constrained flow (13) under the hypotheses of Charous and Lermusiaux [14, Lemma 2.5.2] applied to the deflated problem. The accuracy of the constrained flow with respect to the full flow is the subject of Section 5.

4.6 Interpretation

The tiers form a continuum between two classical methods. With $\mathcal{A} = \emptyset$, $\mathcal{B} = [r]$, all modes are Galerkin modes, the basis U is frozen and (25) is a POD–Galerkin model whose basis is the DO subspace at the last activation; the defect is $\|P_U^\perp F\|^2$, the full normal component. With $\mathcal{A} = \mathcal{B} = [r]$ the scheme is the rank-adaptive DORK method and the defect is $\|P_U^\perp F P_Z^\perp\|^2$. In between, the scheduler buys down the defect one mode at a time at a marginal price of s_j^2 per rotated mode and c_j^2 per advanced coefficient. Because freezing does not alter the state $\tilde{X}$ but only its velocity, switching a mode between tiers is continuous in the state: the scheme is a switched system whose error is governed by the time-integrated defects, as analyzed next. From the multirate viewpoint [32, 37], a frozen mode is a slow component advanced with an infinite step; a mode that is reactivated at every K th scheduling interval is a slow component advanced with the macro-step $K\Delta t$, and Lemma 5.7 below plays the role of the interpolation error of the slow components.

5 Error analysis

We now quantify the error of the selective scheme with respect to the full-rank solution of (1). The analysis proceeds in three layers: the continuous selective flow (13) versus the exact flow (Theorem 5.5(i)), the discrete selective step versus the discrete full step and versus the continuous selective flow (Theorem 5.4, Theorem 5.5(ii)), and the effect of holding the selection fixed over a scheduling interval (Lemma 5.7, Corollary 5.8). Throughout, $\tilde{X}(t)$ denotes the selective approximation, $X(t)$ the exact solution, and ℓ a Lipschitz constant of F .

5.1 Assumptions

Assumption 5.1 (Dynamics). F is one-sided Lipschitz in X with constant $\ell \in \mathbb{R}$ on a tube around the trajectories, $\langle F(X, t) - F(Y, t), X - Y \rangle \leq \ell \|X - Y\|^2$, and Lipschitz with constant

$\ell_+ \geq \max(\ell, 0)$; $\partial_t F$ is bounded there.

One-sided Lipschitz constants [18, 39] allow $\ell < 0$ for dissipative dynamics, in which case the bounds below do not grow in time.

Assumption 5.2 (Approximability). Along the trajectory, the Dirac–Frenkel defect is bounded, $\|(I - P_{T_{\tilde{X}}\mathcal{M}_r})F(\tilde{X}, t)\| \leq \varepsilon_N(t)$, and the basis- and coefficient-staleness defects of Theorem 4.4 are bounded by the scheduled values $\|\varepsilon_S(t)\| \leq \varepsilon_S(t)$, $\|\varepsilon_Z(t)\| \leq \varepsilon_Z(t)$.

The first part is the ε -approximability condition of Ceruti and Lubich [10], Kieri et al. [47]; the second is what the scheduler of Section 6 enforces.

Assumption 5.3 (Discrete one-step maps). Let $\mathcal{S}_i$ denote the one-step map of the selective scheme on step i (Runge–Kutta stages of order p evaluated on the low-rank state, retraction of order q , coefficient update (23)). On a neighborhood $\mathcal{N}_i \subset \mathcal{M}_r^{\mathcal{F}_i}$ of the trajectory, $\mathcal{S}_i$ is Lipschitz with constant $1 + L_S \Delta t_i$, and its local error with respect to the continuous selective flow $\varphi_{t_i \rightarrow t_{i+1}}^{\text{sel}}$ satisfies $\|\mathcal{S}_i(Y) - \varphi_{t_i \rightarrow t_{i+1}}^{\text{sel}}(Y)\| \leq C_{RK} \Delta t_i^{p+1} + C_{ret} \Delta t_i^{q+1}$ for $Y \in \mathcal{N}_i$.

The consistency part is Charous and Lermusiaux [14, Lemma 2.5.2] transferred to the deflated problem by Theorem 4.7; the Lipschitz part is the standard stability of projected Runge–Kutta methods [40, 46]. As in all DLRA analyses based on the tangent projection [30, 48], L_S and C_{ret} depend on the curvature of $\mathcal{M}_r$ through the inverse of the smallest *active* singular value, $\sigma_{\min, \mathcal{A}} = \min_{j \in \mathcal{A}} \sigma_j$, unless robust retractions (24) are used; the frozen modes do not enter these constants (Proposition 5.9(c)).

5.2 Local error of the selective step

Theorem 5.4 (Selective versus full step). Let $\tilde{X}_i = U_i Z_i^\top$, $\chi = \tilde{X}_i + \Delta t \Phi$, and let $\tilde{X}_{i+1}^{\text{full}}$ and $\tilde{X}_{i+1}^{\text{sel}}$ be the results of the full DORK step and of the selective step (23) with the same integrated operator Φ , the same retraction and $\mathcal{D} \subseteq \mathcal{F}$. Let $\varepsilon_{S,i} = P_{U_i}^\perp \Phi(P_{Z_i} - P_{Z_{i,\mathcal{A}}})$ and $\varepsilon_{Z,i} = U_{i,\mathcal{D}} U_{i,\mathcal{D}}^\top \Phi$ be the staleness defects of Theorem 4.4 evaluated with Φ in place of F . Then

$$\begin{aligned} \tilde{X}_{i+1}^{\text{sel}} - \tilde{X}_{i+1}^{\text{full}} &= -\Delta t (\varepsilon_{S,i} + \varepsilon_{Z,i}) + \mathcal{O}(\Delta t^2), \\ \|\tilde{X}_{i+1}^{\text{sel}} - \tilde{X}_{i+1}^{\text{full}}\| &\leq \Delta t \sqrt{\|\varepsilon_{S,i}\|^2 + \|\varepsilon_{Z,i}\|^2} + C_i \Delta t^2, \end{aligned} \tag{27}$$

with $C_i \leq \kappa (\|\Phi\|^2 / \sigma_{\min,i} + \|\Phi\|^2 \|\chi\| / \sigma_{\min,i}^2)$ for an absolute constant κ , where $\sigma_{\min,i}$ is the smallest singular value of $\tilde{X}_i$. The leading term is independent of the order of the retraction.

The proof is in Appendix A.1. The two leading contributions are orthogonal (their column spaces lie in $\text{range } U_i^\perp$ and $\text{range } U_{i,\mathcal{D}}$, respectively), which is why they add in quadrature; the discrete per-step difference between selective and full updates is thus Δt times the instantaneous staleness defect, exactly as Theorem 4.4 predicts for the continuous flows. In numerical tests with random (U, Z, Φ) the ratio of the left-hand side of (27) to $\Delta t \sqrt{\|\varepsilon_{S,i}\|^2 + \|\varepsilon_{Z,i}\|^2}$ is $1 - \mathcal{O}(\Delta t)$ (Section 9).

5.3 Global error

Theorem 5.5 (Global error of the selective scheme). Let Assumptions 5.1 and 5.2 hold.

(i) (Continuous time.) The solution $\tilde{X}(t)$ of the selective DO equations (13), with piecewise-constant $(\mathcal{A}, \mathcal{B}, r)$ and continuous $\tilde{X}$ across switches, satisfies

$$\|\tilde{X}(t) - X(t)\| \leq e^{\ell t} \|\tilde{X}(0) - X(0)\| + \int_0^t e^{\ell(t-s)} (\varepsilon_N(s) + \varepsilon_S(s) + \varepsilon_Z(s)) ds. \quad (28)$$

(ii) (Discrete time.) Let in addition Assumption 5.3 hold, and let $\tilde{X}_N$ be the result of N selective steps with step sizes Δt_i , $t_{i+1} = t_i + \Delta t_i$, $t_N = T$. Then

$$\begin{aligned} \|\tilde{X}_N - X(T)\| &\leq e^{\ell T} \|\tilde{X}_0 - X(0)\| + \sum_{i=0}^{N-1} \omega_i \Delta t_i (\varepsilon_{N,i} + \varepsilon_{S,i} + \varepsilon_{Z,i}) \\ &\quad + \sum_{i=0}^{N-1} e^{L_S(T-t_{i+1})} (C_{RK} \Delta t_i^{p+1} + C_{ret} \Delta t_i^{q+1}), \quad \omega_i = \frac{1}{\Delta t_i} \int_{t_i}^{t_{i+1}} e^{\ell(T-s)} ds, \end{aligned} \quad (29)$$

where $\varepsilon_{\cdot,i} = \sup_{[t_i, t_{i+1}]} \varepsilon_{\cdot}(t)$ and $\omega_i \leq e^{\ell(T-t_{i+1})} \max(1, e^{\ell \Delta t_i})$.

The proof (Appendix A.2) combines the defect decomposition with Gronwall's inequality [36] for (i) and with the ‘‘Lady Windermere’s fan’’ argument of Hairer et al. [40, Sec. II.3] for (ii). Theorem 5.5 makes precise the sense in which the selective scheme is a *controlled* approximation of the full DORK scheme: relative to Kieri et al. [47], Koch and Lubich [48], the only new terms are the time-integrated staleness defects, which the scheduler keeps below a prescribed budget, and which vanish as $\mathcal{A}, \mathcal{B} \rightarrow [r]$.

Corollary 5.6 (Error-rate functional). *Define the error rate of step i as*

$$\rho_i(\Delta t_i, r_i, \mathcal{A}_i, \mathcal{B}_i) = \varepsilon_{N,i}(r_i) + \varepsilon_{S,i}(\mathcal{A}_i) + \varepsilon_{Z,i}(\mathcal{B}_i) + C_{RK} \Delta t_i^p + C_{ret} \Delta t_i^q. \quad (30)$$

For $\ell \geq L_S$ the bound (29) reads $\|\tilde{X}_N - X(T)\| \leq e^{\ell T} \|\tilde{X}_0 - X(0)\| + \sum_i \omega'_i \Delta t_i \rho_i$ with $\omega'_i = \max(\omega_i, e^{L_S(T-t_{i+1})})$; since the weight ω'_i is common to all admissible choices at step i , minimizing the bound over the sequence of decisions decouples into minimizing $\Delta t_i \rho_i$ per unit of physical time, i.e., ρ_i , at each scheduling instant subject to the cost constraints of Section 6.

Proof. Bound both weights in (29) by ω'_i and factor Δt_i from each summand. $\square$

When $L_S > \ell$ the two sums carry different weights, and the scheduler uses ρ_i with the discretization terms multiplied by $e^{(L_S-\ell)(T-t_{i+1})}$; in practice both constants are estimated a posteriori (Section 6.2) and the difference is immaterial.

5.4 Fixed-interval scheduling: drift of the scores

The selection $(\mathcal{A}, \mathcal{B})$ is computed from the scores at a scheduling instant t_k and held over the interval $[t_k, t_k + \tau]$. The following lemma bounds the resulting loss of optimality by the drift of the score vectors.

Lemma 5.7 (Score drift). *Let $s(t) = (s_j(t))_{j \in [r]}$ and $c(t) = (c_j(t))_{j \in [r]}$ be the score vectors of (17) along the trajectory, and let L_s, L_c bound their time derivatives in the Euclidean norm, $\|\dot{s}(t)\|_2 \leq L_s$, $\|\dot{c}(t)\|_2 \leq L_c$. Let $\mathcal{A}_k$ minimize $\varepsilon_S(t_k; \mathcal{A}) = (\sum_{j \notin \mathcal{A}} s_j(t_k)^2)^{1/2}$ over $|\mathcal{A}| = a$, and let $\varepsilon_S^*(t) = \min_{|\mathcal{A}|=a} \varepsilon_S(t; \mathcal{A})$. Then for $t \in [t_k, t_k + \tau]$,*

$$\varepsilon_S(t; \mathcal{A}_k) \leq \varepsilon_S^*(t) + 2L_s(t - t_k) \leq \varepsilon_S^*(t) + 2L_s\tau, \quad (31)$$

and likewise $\varepsilon_Z(t; \mathcal{B}_k) \leq \varepsilon_Z^*(t) + 2L_c(t - t_k)$. Under Assumption 5.1 and in the KL gauge, with $\|F\| \leq F_{\max}$, $\|\partial_t F\| \leq F'_{\max}$ and $\sigma_j \geq \sigma_{\min}$,

$$L_s \leq \sqrt{r} \left(\ell_+ F_{\max} + F'_{\max} + \frac{4F_{\max}^2}{\sigma_{\min}} \right), \quad L_c \leq \sqrt{r} \left(\ell_+ F_{\max} + F'_{\max} + \frac{F_{\max}^2}{\sigma_{\min}} \right). \quad (32)$$

Proof. For any fixed $\mathcal{A}$, $\varepsilon_S(t; \mathcal{A}) = \|s_{\mathcal{F}}(t)\|_2$ with $\mathcal{F} = [r] \setminus \mathcal{A}$, so

$$|\varepsilon_S(t; \mathcal{A}) - \varepsilon_S(t_k; \mathcal{A})| \leq \|s_{\mathcal{F}}(t) - s_{\mathcal{F}}(t_k)\|_2 \leq \|s(t) - s(t_k)\|_2 \leq L_s(t - t_k).$$

Hence $\varepsilon_S(t; \mathcal{A}_k) \leq \varepsilon_S(t_k; \mathcal{A}_k) + L_s(t - t_k) = \varepsilon_S^*(t_k) + L_s(t - t_k)$, and, with $\mathcal{A}^*(t)$ a minimizer at time t ,

$$\varepsilon_S^*(t_k) \leq \varepsilon_S(t_k; \mathcal{A}^*(t)) \leq \varepsilon_S(t; \mathcal{A}^*(t)) + L_s(t - t_k) = \varepsilon_S^*(t) + L_s(t - t_k).$$

The bounds (32) are derived in Appendix A.3 by differentiating $s_j = \|P_U^\perp F z_j\| / \sigma_j$ and $c_j = \|F^\top u_j\|$ along the flow and using $\|\dot{U}\| \leq \|F\| / \sigma_{\min}$, $\|\dot{Z}\| \leq \|F\|$, $|\dot{\sigma}_j| \leq \|F\|$ and $\|dF/dt\| \leq \ell_+ \|F\| + \|\partial_t F\|$. $\square$

The bound (32) is a worst case; in practice L_s and L_c are estimated from finite differences of the score vectors at consecutive scheduling instants, and (31) then provides the rule $\tau \lesssim \delta / (2L_s)$ for the interval length that keeps the scheduling loss below a fraction δ of the error budget. Inserted in Theorem 5.5, the scheduling loss adds at most $2(L_s + L_c)\tau(e^{\ell T} - 1)/\ell$ to the global error.

Corollary 5.8 (Activation certificate; lazy score refresh). *Let θ_s be the activation threshold implied by the selection at t_k (the smallest active score, $\theta_s = \min_{j \in \mathcal{A}_k} s_j(t_k)$) and let $L_{s,\max} \geq \sup_{t,j} |\dot{s}_j(t)|$. A frozen mode $j \in \mathcal{F}_k$ cannot overtake an active one before*

$$t_j^{\text{cert}} = t_k + \frac{\theta_s - s_j(t_k)}{2L_{s,\max}}, \quad (33)$$

so its score need not be recomputed before t_j^{cert} , and the scheduling interval may be taken as $\tau_k = \min(\tau_{\max}, \min_{j \in \mathcal{F}_k} (t_j^{\text{cert}} - t_k))$. The same holds for dormant coefficients with $\theta_c = \min_{j \in \mathcal{B}_k} c_j(t_k)$ and $L_{c,\max}$.

Proof. By the mean value theorem, $s_j(t) \leq s_j(t_k) + L_{s,\max}(t - t_k)$ and $\min_{i \in \mathcal{A}_k} s_i(t) \geq \theta_s - L_{s,\max}(t - t_k)$; the two cannot cross before $t_k + (\theta_s - s_j(t_k)) / (2L_{s,\max})$. $\square$

This is the lazy-greedy mechanism of Krause and Golovin [49], Minoux [57]: a stale score is a certified bound on the current one, and the scheduler only pays for recomputing the scores of modes whose certificates have expired. Since computing the score of a frozen mode costs as much as its update (both require $P_U^\perp F z_j$), the certificates are what make the scheduler's overhead sublinear in r (Section 6.1).

5.5 Interplay with rank adaptation

Proposition 5.9 (Rank changes and frozen modes). *(a) (Augmentation.) Let the rank be increased from r to $r+k$ by appending $Q \in \mathcal{V}_{m,k}$, $Q^\top U = 0$, with zero coefficients, followed by the kick $\hat{Z} \leftarrow \hat{Z} + \Delta t \Phi^\top \hat{U}$ [14, Eq. (2.26)]. The new singular values satisfy $\sigma_{r+1}, \dots, \sigma_{r+k} \leq \Delta t \|\Phi\|$ (Weyl's inequality, 80), their scores $s_j = \|P_U^\perp \Phi z_j\| / \sigma_j$ are undefined at insertion (0/0) and equal $\|P_U^\perp \Phi \Phi^\top q_j\| / \|\Phi^\top q_j\|$ after the kick; hence new modes must be active, with the robust retraction (24), for at least one scheduling interval before they become eligible for freezing.*

- (b) (Truncation.) In the KL gauge, removing mode j changes $\tilde{X}$ by $\|u_j z_j^\top\| = \sigma_j$ exactly, so a dormant mode ($c_j \approx 0$) with σ_j below the truncation tolerance is removed at a one-off error σ_j , which the scheduler compares to the cost of carrying it.
- (c) (Conditioning.) The selective mode equation inverts only the active covariance $Z_{\mathcal{A}}^\top Z_{\mathcal{A}}$, a principal submatrix of $Z^\top Z$; by Cauchy interlacing [34, Thm. 8.1.7] its eigenvalues lie in $[\lambda_{\min}(Z^\top Z), \lambda_{\max}(Z^\top Z)]$, so $\kappa(Z_{\mathcal{A}}^\top Z_{\mathcal{A}}) \leq \kappa(Z^\top Z)$, with equality only if $\mathcal{A}$ contains both the largest and the smallest mode. Freezing weak modes therefore regularizes the DO covariance inversion, and the curvature constants of Assumption 5.3 involve $\sigma_{\min, \mathcal{A}}$ rather than $\sigma_{\min}$.

Proof. (a) $\tilde{X}_{i+1}$ is a perturbation of $\tilde{X}_i$ (rank r) by $\Delta t \Phi$, so by Weyl's inequality for singular values $\sigma_{r+1}(\tilde{X}_{i+1}) \leq \sigma_{r+1}(\tilde{X}_i) + \Delta t \|\Phi\| = \Delta t \|\Phi\|$; after the kick, $z_{r+1} = \Delta t \Phi^\top q_1$ and $\sigma_{r+1} = \Delta t \|\Phi^\top q_1\|$, giving the stated score. (b) In the KL gauge the $u_j z_j^\top$ are mutually orthogonal rank-one matrices of norm σ_j . (c) Interlacing for principal submatrices of symmetric matrices. $\square$

Part (c) is a side benefit of selective updates that is worth emphasizing: the covariance inversion that makes the original DO equations stiff [47, 62] is confined to the active block, and the weakest modes, the ones responsible for the ill-conditioning, are also the ones with the smallest scores $s_j = \sigma_j \|\dot{u}_j\|$ whenever their rotation rate is not anomalously large.

6 Cost model and joint Pareto scheduling

The error analysis of Section 5 supplies an error-rate functional ρ (Corollary 5.6) whose ingredients are computable; this section supplies the matching cost model and turns the joint choice of $(\Delta t, r, \mathcal{A}, \mathcal{B})$ into a multiobjective problem that is solved exactly by enumeration at each scheduling instant.

6.1 Cost model

We write the cost of one selective step as

$$C_{\text{step}}(r, a, b) = C_0(r) + a C_U(r) + b C_Z(r) + \frac{C_{\text{sch}}(r)}{K} + C_{\text{sw}} n_{\text{sw}}, \quad (34)$$

where C_0 collects the work that does not depend on the active sets (mean update, evaluation of the sampled or contracted right-hand side, embedded error estimate), C_U is the cost of advancing one active basis vector (its DO right-hand side, its share of the retraction and of the re-orthonormalization), C_Z the cost of advancing one coefficient column, C_{sch} the cost of one scheduling decision amortized over the K steps of the interval, and C_{sw} a switching cost charged per tier change (n_{sw} changes) to discourage chattering [cf. the memory/forgetting rates used for rank changes in 14]. Table 3 gives leading-order operation counts for the two evaluation strategies of the right-hand side that arise in practice. For a quadratic nonlinearity $F(X) = AX + Q(X, X)$ (incompressible Navier–Stokes, quasi-geostrophy) the DO right-hand sides are assembled by contraction of the pairwise products $Q(u_k, u_l)$ with the third moments $\mathbb{E}[Y_j Y_k Y_l]$ [68, 77]; for a general nonlinearity (compressible Navier–Stokes, Section 8) the right-hand side is evaluated at q sampled realizations and p sampled spatial points and reconstructed by a CUR/oblique projection [24, 61]. In both cases the quantities that scale *quadratically* with the rank in the full scheme, $\mathcal{O}(r^2 m)$ contractions and projections, $\mathcal{O}(r^2)$ coefficient couplings, scale as $\mathcal{O}(r(a+b)m)$ in the selective scheme, which is the source of the savings, while the shared cost $C_0(r)$ is unchanged.

The scheduler's own cost deserves comment. Computing the score s_j of a *frozen* mode requires $P_j^\top \Phi z_j$, i.e., the same work as updating it. Without further measures the scheduler would therefore

Table 3: Leading-order operation counts per step. C_m : cost of one deterministic right-hand-side evaluation (or implicit solve) on m degrees of freedom; c_Q : per-point cost of the quadratic product; q, p : numbers of sampled realizations and spatial points ($q \gtrsim r$); k : number of candidate augmentation vectors; K : steps per scheduling interval. Constants are calibrated online from measured timings (Section 6.2).

Term	Galerkin contraction (quadratic F)	Sampled/CUR evaluation (general F)
$C_0(r)$	$C_m + \frac{1}{2}r^2m c_Q + nr^3$	$qC_m + np c_F + q^2p$
$C_U(r)$ (per active mode)	$C_m^{\text{lin}} + r^2m + rm$	$mq + np + rm$
$C_Z(r)$ (per active coefficient)	$r^2m + nr^2$	$mq + np$
retraction, order q_{ret} (per active mode)	$q_{\text{ret}} rm$	$q_{\text{ret}} rm$
$C_{\text{sch}}(r)$ (per interval)	$r C_U(r) + mnk$	$r C_U(r) + m q k$
full DORK step (reference)	$C_0 + r(C_U + C_Z)$	$C_0 + r(C_U + C_Z)$
Monte Carlo step (reference)	nC_m	nC_m

cost $r C_U$ per decision, and the savings of freezing would be lost unless $K \gg r/a$. Two mechanisms keep the overhead small: the activation certificates of Corollary 5.8, which exempt a frozen mode from score recomputation until its certificate expires, and the fact that in the sampled/CUR strategy the products ΦZ and $\Phi^\top U$ for *all* r columns cost $\mathcal{O}(mqr)$ once the sampled columns $\Phi(:, \mathcal{J})$ are available, so that a full score refresh costs one $\mathcal{O}(mqr)$ matrix product rather than r mode updates. In either case the per-step overhead is C_{sch}/K and is included in (34).

6.2 Error-rate model and its estimation

The error-rate functional (30) is evaluated for candidate decisions $(\Delta t', r', \mathcal{A}', \mathcal{B}')$ from quantities available at the scheduling instant t_k (current step size Δt_k , rank r , KL gauge):

1. *Normal defect versus rank.* Let $R = P_U^\perp \Phi P_Z^\perp$ be the residual (the Dirac–Frenkel defect with Φ in place of F) and $\varrho_1 \geq \varrho_2 \geq \dots$ its singular values, computed for the leading k by the randomized SVD [41] exactly as in the rank-augmentation step of Charous and Lermusiaux [14]. Appending the leading k' left singular vectors removes the corresponding energy from the defect, so

$$\varepsilon_N(r + k') \approx \left(\|R\|^2 - \sum_{i \leq k'} \varrho_i^2 \right)^{1/2}, \quad k' = 0, \dots, k. \quad (35)$$

Justification. R is the part of Φ orthogonal to the tangent space; after augmentation by $Q = [\varrho\text{-vectors}]$, the new tangent space contains $Q(\cdot)^\top$ and the remaining normal component is $P_Q^\perp R$, whose norm is the stated tail sum [27].

2. *Truncation.* Removing mode j (rank $r - 1$) adds its entire dynamics to the defect and incurs the one-off error σ_j (Proposition 5.9(b)); over an interval of length τ we charge

$$\varepsilon_{\text{trunc}}(j) = (s_j^2 + c_j^2)^{1/2} + \sigma_j/\tau, \quad (36)$$

because a removed mode is a mode that is both frozen and dormant (Theorem 4.4) and absent from the state.

3. *Staleness.* From the sorted scores, $\varepsilon_S(a') = (\sum_{j > a'} s_{(j)}^2)^{1/2}$ and $\varepsilon_Z(b') = (\sum_{j > b'} c_{(j)}^2)^{1/2}$ with $s_{(1)} \geq s_{(2)} \geq \dots$ and $c_{(1)} \geq c_{(2)} \geq \dots$ (Lemma 4.6); by Lemma 5.7 a margin $2(L_s + L_c)\tau$ is added, with L_s, L_c estimated from finite differences of the score vectors over the previous interval.

4. *Time integration.* The perturbative expansion $\Delta t \Phi = \sum_j \Delta t^j \Phi_j$ of the DORK schemes provides the embedded estimate $\hat{e}_k = \|\Phi^{(p)} - \Phi^{(p-1)}\| \Delta t_k = \|\Phi_p\| \Delta t_k^p$ of the local error of the order- $(p-1)$ companion [40, Sec. II.4]; extrapolating with the standard asymptotic model gives the rate

$$\varepsilon_{RK}(\Delta t') = \frac{\hat{e}_k}{\Delta t_k} \left(\frac{\Delta t'}{\Delta t_k} \right)^{p-1}. \quad (37)$$

Justification. Local error $\propto \Delta t^p$ for the embedded method; dividing by $\Delta t'$ gives the rate [40, 73].

5. *Retraction.* The automatic gradient-descent retraction [14, Alg. 2.2] returns the norm Δ of the last iterate correction, an estimate of $\varepsilon_{pr} = \mathcal{O}(\Delta t^{q+1})$; we use $\varepsilon_{ret}(\Delta t') = (\Delta/\Delta t_k)(\Delta t'/\Delta t_k)^q$, and set $\varepsilon_{ret} = 0$ when the retraction is iterated to tolerance.
6. *Stability constraint.* Explicit stages impose $\Delta t' \leq \Delta t_{\text{stab}}$ (CFL-type, Section 8); this is a hard constraint on the decision set rather than an error term.

The combined error rate of a candidate is

$$\rho(\Delta t', r', a', b') = \varepsilon_N(r') + \varepsilon_S(a') + \varepsilon_Z(b') + \varepsilon_{RK}(\Delta t') + \varepsilon_{ret}(\Delta t') + 2(L_s + L_c)\tau, \quad (38)$$

and Theorem 5.5 guarantees that keeping $\rho \leq \text{tol}$ on every interval yields $\|\tilde{X}_N - X(T)\| \leq e^{\ell T} \|\tilde{X}_0 - X(0)\| + \text{tol}(e^{\ell T} - 1)/\ell$ (with the usual replacement $(e^{\ell T} - 1)/\ell \rightarrow T$ when $\ell = 0$).

6.3 The multiobjective problem

At a scheduling instant the decision set is

$$\Xi = \left\{ (\Delta t', r', a', b') : \Delta t' \in [\Delta t_{\min}, \min(\Delta t_{\max}, \Delta t_{\text{stab}})], r' \in \{r-k_-, \dots, r+k_+\}, 0 \leq a' \leq b' \leq r' \right\}, \quad (39)$$

with the sets $\mathcal{A}', \mathcal{B}'$ determined from (a', b') by Lemma 4.6 (new modes forced active, Proposition 5.9(a); truncated modes taken from the dormant set with smallest σ_j). The two objectives are the *cost rate* and the *error rate*,

$$\mathcal{J}_C(\xi) = \frac{\mathcal{C}_{\text{step}}(r', a', b')}{\Delta t'}, \quad \mathcal{J}_E(\xi) = \rho(\Delta t', r', a', b'), \quad \xi \in \Xi. \quad (40)$$

A decision ξ is *Pareto optimal* if no ξ' satisfies $\mathcal{J}_C(\xi') \leq \mathcal{J}_C(\xi)$ and $\mathcal{J}_E(\xi') \leq \mathcal{J}_E(\xi)$ with at least one strict inequality [33, 63]; the set of Pareto-optimal decisions is the *front* $\Pi \subset \Xi$.

Proposition 6.1 (Exact enumeration). *Discretize $\Delta t'$ on a geometric grid of G points. Then $|\Xi| \leq (k_- + k_+ + 1)(r + k_+ + 1)^2 G$, every $(\mathcal{J}_C, \mathcal{J}_E)$ is a closed-form expression in the sorted scores, the residual spectrum, the embedded estimates and the calibrated costs, and the front Π is obtained by sorting the candidates by $\mathcal{J}_C$ and retaining those with a running minimum of $\mathcal{J}_E$. The cost of the scheduler is therefore $\mathcal{O}(|\Xi| \log |\Xi|)$ scalar operations, independent of m and n , in addition to the score refresh and residual SVD of Table 3.*

Proof. Immediate from (34), (38) and Lemma 4.6; the sweep is the standard $\mathcal{O}(N \log N)$ non-dominated sort in two dimensions [22]. $\square$

For $r = 50$, $k_{\pm} = 2$ and $G = 16$ the enumeration has about 2×10^5 candidates and takes milliseconds; it is negligible compared with a single field evaluation.

Selecting a point on the front. The front Π is reduced to a single decision by one of the following rules, which are the standard choices of a posteriori multi-objective decision making [54, 56]:

- (R1) *Error budget.* $\xi^* = \arg \min\{\mathcal{J}_C(\xi) : \xi \in \Pi, \mathcal{J}_E(\xi) \leq \text{tol}\}$. This is the direct analogue of tolerance-based step-size control [40, 73] and carries the guarantee of Theorem 5.5. If the budget cannot be met (typically because the rank cap $r_{\max}$ binds), the cheapest decision within a factor $1 + \delta_f$ (we use $\delta_f = 0.1$) of the minimal achievable error rate is taken and the event is flagged, so that no cost is spent on decisions that reduce the error only marginally.
- (R2) *Knee.* Normalize both objectives to $[0, 1]$ over Π and select the point of maximal perpendicular distance from the chord joining the two anchor points (minimum-cost and minimum-error decisions) [8, 21]. This rule needs no tolerance and identifies the decision beyond which further error reduction becomes disproportionately expensive; because both objectives span several decades, the distance is measured in logarithmic coordinates, and a hysteresis band (the current decision is kept if it lies within a factor $1 + \delta_h$ of the knee in both objectives) limits chattering.
- (R3) *Cost budget.* $\xi^* = \arg \min\{\mathcal{J}_E(\xi) : \xi \in \Pi, \mathcal{J}_C(\xi) \leq C_{\max}\}$ for a fixed compute budget per unit time.

Weighted-sum scalarization is avoided because the front is discrete and typically nonconvex, so weighted sums can miss Pareto points [20]; the three rules above do not have this limitation.

6.4 Scheduling intervals and emergency re-scheduling

Decisions are taken at instants t_k and held for K_k steps, $\tau_k = K_k \Delta t_k$. The interval is the minimum of a user maximum $\tau_{\max}$, the certificate bound of Corollary 5.8, and the drift bound $\delta \text{tol} / (2(L_s + L_c))$ of Lemma 5.7 for a chosen fraction δ of the budget. Within an interval three events trigger an immediate re-scheduling: (i) the embedded estimate $\hat{e}_i$ exceeds twice its budgeted value (the classical rejection test, 40, 74); (ii) the angle criterion $\theta > \theta^*$ of Charous and Lermusiaux [14] fires on the active block, signaling that the rank must grow; (iii) a certificate expires. Rank increases are executed as in Charous and Lermusiaux [14, Alg. 2.3] with the robust selective retraction (24), and the KL rotation that accompanies rank truncation also resets the gauge (Remark 4.2).

6.5 Consistency and limiting behavior

Proposition 6.2 (Recovery of full DORK and of fixed-basis Galerkin models). *Under rule (R1), as $\text{tol} \rightarrow 0$ the selected decisions satisfy $\mathcal{A}^* = \mathcal{B}^* = [r^*]$ and $\Delta t^* \rightarrow 0$; the scheme reduces to the rank-adaptive so-/gd-DORK method of Charous and Lermusiaux [14]. Under rule (R3), as $C_{\max} \rightarrow C_0(r)/\Delta t_{\text{stab}}$ from above, the selected decisions satisfy $a^* = 0$ and the scheme reduces to a POD-Galerkin model on the frozen DO basis with coefficients advanced by (25).*

Proof. Every term of (38) is nonnegative and $\varepsilon_S(a') > 0$ whenever $a' < r'$ and some frozen score is positive, and similarly for ε_Z ; hence feasibility of $\rho \leq \text{tol}$ for tol below the smallest positive score forces $a' = b' = r'$, and $\varepsilon_{RK}(\Delta t') \leq \text{tol}$ forces $\Delta t' \rightarrow 0$. The second statement follows because $C_{\text{step}} \geq C_0$ with equality iff $a' = b' = 0$, and the cheapest feasible decision with $b' = r'$, $a' = 0$ is preferred to $a' = b' = 0$ whenever C_Z is affordable, which Lemma 4.8(a) also recommends. $\square$

7 Algorithms

Algorithms 1 to 3 summarize the method. Algorithm 1 is one selective gd-DORK step; it calls the retractions of Charous and Lermusiaux [14] on the deflated active block (Theorem 4.7) and returns the embedded and retraction error estimates of Section 6.2. Algorithm 2 is the Pareto decision of Section 6.3, and Algorithm 3 the time loop with scheduling intervals, certificates and emergency triggers. The so-DORK variant replaces lines 5–8 of Algorithm 1 by the selective so-DORK corrections (Section 4.5).

Input: $U_i \in \mathcal{V}_{m,r}$, $Z_i \in \mathbb{R}_*^{n \times r}$, Δt , sets $\mathcal{A}, \mathcal{B}$, Runge–Kutta stages providing $\Phi^{(1)}, \dots, \Phi^{(k)}$ (partial sums of $\Delta t \Phi = \sum_j \Delta t^j \Phi_j$), retraction $\mathcal{R}$ (optimal perturbative, robust, or automatic gradient-descent)

Output: U_{i+1}, Z_{i+1} ; embedded estimate $\hat{e}$; retraction estimate Δ

$\mathcal{F} \leftarrow [r] \setminus \mathcal{A}$, $\mathcal{D} \leftarrow [r] \setminus \mathcal{B}$; $\tilde{X}_i \leftarrow U_i Z_i^\top$

Evaluate the Runge–Kutta stages on the low-rank state (all tiers present) to obtain $\Phi^{(1)}, \dots, \Phi^{(k)}$ in factored or sampled form (Section 8.4)

$\hat{e} \leftarrow \Delta t \|\Phi^{(k)} - \Phi^{(k-1)}\|$ *embedded error estimate, (37)* $U' \leftarrow U_{i,\mathcal{A}}$, $Z' \leftarrow Z_{i,\mathcal{A}}$, $\tilde{X}' \leftarrow U' Z'^\top$
deflated active block

for $j = 1, \dots, k$ **do**

$\Phi' \leftarrow P_{U_{i,\mathcal{F}}}^\perp \Phi^{(j)}$ *deflation, (20)* $U' Z'^\top \leftarrow \mathcal{R}_{U' Z'^\top}(\Delta t \Phi' - (U' Z'^\top - \tilde{X}'))$ with U'

re-orthonormalized against $U_{i,\mathcal{F}}$ Charous and Lermusiaux [14, Alg. 2.5] on the deflated problem

end for

$\Delta \leftarrow$ norm of the last gradient-descent correction (or 0 for a fixed-order retraction)

$U_{i+1} \leftarrow [U_{i,\mathcal{F}} \ U']$ (original column order)

$\chi \leftarrow \tilde{X}_i + \Delta t \Phi^{(k)}$; $Z_{i+1,\mathcal{B}} \leftarrow \chi^\top U_{i+1,\mathcal{B}}$; $Z_{i+1,\mathcal{D}} \leftarrow Z_{i,\mathcal{D}}$ (23); Galerkin modes obey (25)

return $U_{i+1}, Z_{i+1}, \hat{e}, \Delta$

Algorithm 1: Selective gd-DORK step of order k (basis-active set $\mathcal{A}$, coefficient-active set $\mathcal{B} \supseteq \mathcal{A}$)

8 Application to stochastic three-dimensional compressible Navier–Stokes flows

8.1 Governing equations

We consider the compressible Navier–Stokes equations for an ideal gas in conservative form [3, 81],

$$\partial_t \rho + \nabla \cdot (\rho \mathbf{u}) = 0, \quad \partial_t (\rho \mathbf{u}) + \nabla \cdot (\rho \mathbf{u} \otimes \mathbf{u} + p \mathbf{I} - \boldsymbol{\tau}) = 0, \quad \partial_t (\rho E) + \nabla \cdot ((\rho E + p) \mathbf{u} - \boldsymbol{\tau} \cdot \mathbf{u} + \mathbf{q}_h) = 0, \quad (41)$$

with $p = (\gamma - 1)(\rho E - \frac{1}{2} \rho |\mathbf{u}|^2)$, $\boldsymbol{\tau} = \mu(\nabla \mathbf{u} + \nabla \mathbf{u}^\top - \frac{2}{3}(\nabla \cdot \mathbf{u})\mathbf{I})$, $\mathbf{q}_h = -\kappa \nabla T$, $T = p/(\rho R_g)$, and a temperature-dependent viscosity (Sutherland’s law, 76, or a power law) with constant Prandtl number. After non-dimensionalization by reference density, speed of sound and length, the parameters are the Reynolds number Re , the Mach number Ma , the Prandtl number Pr and γ . The state per realization is $\mathbf{w} = (\rho, \rho u, \rho v, \rho w, \rho E)$ on N_x cells or nodes, so $m = 5N_x$; the right-hand side F is the spatial discretization of the divergence terms, a *non-polynomial* function of $\mathbf{w}$ because of the division by ρ in the velocity, the pressure–energy relation and the temperature-dependent transport coefficients. This distinguishes the proposed formulation from the incompressible and quasi-geostrophic DO applications [70, 77], whose nonlinearities are quadratic and admit exact Galerkin contraction.

Input: U, Z (rotated to the KL gauge), Φ (sampled/factored), current $\Delta t_k, \hat{e}_k, \Delta_k$; calibrated costs $C_0, C_U, C_Z, C_{\text{sch}}, C_{\text{sw}}$; rule (R1)/(R2)/(R3) with tol or C_{max} ; bounds $k_{\pm}, G, \Delta t_{\min}, \Delta t_{\max}, \Delta t_{\text{stab}}, \tau_{\max}$; previous scores for drift estimates

Output: $\Delta t_{k+1}, r_{k+1}, \mathcal{A}_{k+1}, \mathcal{B}_{k+1}$, interval length τ_{k+1}

Scores: $s_j \leftarrow \|P_U^\perp \Phi z_j\| / \sigma_j, c_j \leftarrow \|\Phi^\top u_j\|$ for modes whose certificate has expired (Corollary 5.8); keep certified stale values otherwise

Residual spectrum: $\varrho_1, \dots, \varrho_{k_+}$ and vectors Q of $R = P_U^\perp \Phi P_Z^\perp$ by randomized SVD [41]

Drift constants L_s, L_c from finite differences of the score vectors

Sort s and c ; form $\varepsilon_S(a'), \varepsilon_Z(b')$ for $a', b' = 0, \dots, r'$ (Lemma 4.6); $\varepsilon_N(r')$ from (35)–(36)

for each $r' \in \{r - k_-, \dots, r + k_+\}, a' \leq r', b' \leq r', \Delta t'$ on the geometric grid **do**

$\mathcal{J}_C \leftarrow C_{\text{step}}(r', a', b') / \Delta t'$ by (34), including $C_{\text{sw}} n_{\text{sw}}$

$\mathcal{J}_E \leftarrow \rho(\Delta t', r', a', b')$ by (38)

end for

Front $\Pi \leftarrow$ non-dominated candidates (sort by $\mathcal{J}_C$, running minimum of $\mathcal{J}_E$)

$\xi^* \leftarrow$ decision on Π by rule (R1), (R2) or (R3)

$\tau_{k+1} \leftarrow \min(\tau_{\max}, \min_{j \in \mathcal{F}}(t_j^{\text{cert}} - t_k), \delta \text{tol} / (2(L_s + L_c)))$

if $r_{k+1} > r$ **then** augment with Q and the kick of Charous and Lermusiaux [14, Eq. (2.26)]; force new modes into $\mathcal{A}_{k+1} \cap \mathcal{B}_{k+1}$

if $r_{k+1} < r$ **then** truncate the $r - r_{k+1}$ dormant modes of smallest σ_j and rotate to the KL gauge

return ξ^*, τ_{k+1}

Algorithm 2: Pareto scheduler at instant t_k

Input: U_0, Z_0, T , initial $\Delta t_0, r_0, \mathcal{A}_0 = \mathcal{B}_0 = [r_0]$, scheduler parameters

$t \leftarrow 0, i \leftarrow 0, k \leftarrow 0$; calibrate C_0, C_U, C_Z from the first full steps

while $t < T$ **do**

if $t \geq t_k + \tau_k$ **or** an emergency trigger fired **then** $(\Delta t, r, \mathcal{A}, \mathcal{B}, \tau) \leftarrow$ Algorithm 2; $k \leftarrow k + 1, t_k \leftarrow t$

$(U_{i+1}, Z_{i+1}, \hat{e}, \Delta) \leftarrow$ Algorithm 1($U_i, Z_i, \Delta t, \mathcal{A}, \mathcal{B}$)

if $\hat{e} > 2 \hat{e}_{\text{budget}}$ **then** reject the step, $\Delta t \leftarrow \Delta t / 2$, raise the emergency trigger, **continue**

if angle θ of the active block exceeds θ^* **then** raise the emergency trigger (rank increase)

Update measured timings (exponential moving average) for C_0, C_U, C_Z

$t \leftarrow t + \Delta t, i \leftarrow i + 1$

end while

Algorithm 3: Selective-update rank-adaptive DORK time integration

8.2 Stochastic formulation and low-rank representation

Consider uncertain initial conditions, parameters, forcing, or boundary data parameterized by a random vector $\boldsymbol{\xi}(\omega)$ of dimension d whose n samples $\boldsymbol{\xi}^{(1)}, \dots, \boldsymbol{\xi}^{(n)}$ (Monte Carlo, quasi-Monte Carlo [72] or collocation nodes [82]) define the columns of $X(t) \in \mathbb{R}^{m \times n}$. The DO representation is

$$X(t) \approx \bar{\boldsymbol{w}}(t) \mathbf{1}^\top + U(t) Z(t)^\top, \quad U \in \mathcal{V}_{m,r}, \quad Z \in \mathbb{R}^{n \times r}, \quad \mathbf{1}^\top Z = 0, \quad (42)$$

i.e., a deterministic mean state $\bar{\boldsymbol{w}}$ (always advanced, Remark 4.3) plus r orthonormal spatial modes of the conservative-variable fluctuation with zero-mean coefficient samples; the mean is evolved by the DO mean equation $\dot{\bar{\boldsymbol{w}}} = \frac{1}{n} F(X) \mathbf{1}$ and the fluctuation by the selective equations (15) applied to $F(X) - \dot{\bar{\boldsymbol{w}}} \mathbf{1}^\top$. Whether modes are formed on conservative, primitive or entropy variables is a modeling choice; conservative variables keep the mean equation conservative, whereas entropy or symmetrized variables give an inner product in which the linearized dynamics are energy stable [6, 66]. In the formulation below, we use conservative variables and a weighted inner product.

Inner product. The components of $\boldsymbol{w}$ have different physical dimensions, so the Frobenius norm must be weighted: $\langle A, B \rangle_W = \text{tr}(A^\top W B)$ with $W = \text{diag}(\text{vol}_i \varsigma_\ell^{-2})$, where vol_i is the cell volume and ς_ℓ a reference scale of variable ℓ ($\rho_\infty, \rho_\infty c_\infty, \rho_\infty c_\infty^2$). All results of Sections 4 to 6 hold verbatim in this inner product after the change of variables $X \leftarrow W^{1/2} X$, $F \leftarrow W^{1/2} F$, which maps W -orthonormal bases to orthonormal ones and preserves ranks and singular values [cf. 14, 66].

8.3 Spatial discretization, time integration and stability constraint

Let F denote a spatial discretization of the conservative equations applied to each realization. Its flux, boundary treatment, and spatial order must be specified for a particular numerical study. An explicit Runge–Kutta integrator of order p supplies the stages used in the DORK expansion $\Delta t \Phi = \sum_j \Delta t^j \Phi_j$ [14]. In that setting, the stability constraint is the acoustic-and-viscous CFL condition

$$\Delta t_{\text{stab}} = \text{CFL} \min_{i, \text{realizations}} \left(\frac{|\boldsymbol{u}| + c}{h} + \frac{2\nu}{h^2} \right)^{-1}, \quad (43)$$

evaluated on the sampled realizations of Section 8.4, which enters the decision set (39) as a hard bound. An implicit or IMEX treatment of the viscous terms [4, 45] changes C_0 and C_U (one factorization shared by all fields, Table 3) but not the algorithm.

8.4 Evaluation of the non-polynomial right-hand side

The selective step needs $\Phi^\top U_B$ (coefficient updates), $P_U^\perp \Phi Z_A$ (active modes), the mean increment $\Phi \mathbf{1}/n$, the embedded estimate and, at scheduling instants, the scores and the residual spectrum. Forming Φ column by column costs n evaluations of the deterministic right-hand side, the cost of Monte Carlo; one may use the sampled evaluation of Donello et al. [24], Naderi et al. [61]: select q realizations $\mathcal{J}$ ($q = r + q_o$ with a small oversampling q_o) by a pivoted QR on $Z^\top$ (DEIM/Q-DEIM, 15, 26) and p spatial rows $\mathcal{I}$ by a pivoted QR on U , evaluate the full right-hand side for the q sampled realizations and the right-hand side at the p sampled points for all realizations, and reconstruct

$$\Phi \approx \Phi(:, \mathcal{J}) \Phi(\mathcal{I}, \mathcal{J})^+ \Phi(\mathcal{I}, :), \quad (44)$$

a CUR approximation [35, 53]. All products required by Algorithms 1 and 2 are then formed from the $m \times q$, $q \times p$ and $p \times n$ factors at the costs listed in Table 3. The CUR error $\varepsilon_{\text{CUR}} = \|\Phi - \Phi(:, \mathcal{J}) \Phi(\mathcal{I}, \mathcal{J})^+ \Phi(\mathcal{I}, :)\|$ is an additional defect that enters Theorem 5.5 additively (it is a

perturbation of F); it is bounded in terms of the $(q + 1)$ st singular value of Φ and the pivoting constants [26, 75], can be monitored on held-out columns and added to (38). Because the DO ansatz (42) reconstructs the sampled realizations as $\bar{\mathbf{w}} + U\mathbf{z}_j$, positivity of ρ and p is not guaranteed for realizations far in the tails; a positivity limiter would be required before evaluating F on nonphysical states. The choice of limiter and its effect on the approximation require assessment in a numerical study. Conservation of the mean mass, momentum and energy is inherited from the conservative discretization through the mean equation; the fluctuation modes carry no net mass by construction ($\mathbf{1}^\top \mathbf{Z} = 0$), and full conservation per realization would require the conservative projections of Einkemmer and Joseph [29], Guo and Qiu [38], which we do not pursue here.

8.5 Cost model and expected regime

With C_m the cost of one explicit right-hand-side evaluation on $m = 5N_x$ degrees of freedom, the sampled strategy gives (Table 3) $C_0 \approx qC_m + npc_F$, $C_U \approx C_Z \approx mq + np$ and a full DORK step $\approx qC_m + 2r(mq + np)$, against nC_m for Monte Carlo. Since $C_m \approx c_F m$ with c_F the per-point stencil cost ($\mathcal{O}(10^2)$ flops for a split-form scheme), the ratio of projection to evaluation work in the full scheme is $2rq/(qc_F) = 2r/c_F$: the projections dominate once $r \gtrsim c_F/2$, and the selective scheme reduces them by the factor $2r/(a + b)$. When r is large relative to the stencil cost, selective updates may reduce projection work. At small r , shared evaluation work dominates, so full updates can be preferable.

8.6 Quantities of interest and error metrics

A numerical assessment should report the mean and standard deviation fields of ρ , $\mathbf{u}$ and p ; the ensemble statistics of the volume-averaged kinetic energy $E_k(t) = \frac{1}{2V} \int \rho |\mathbf{u}|^2$, its dissipation rate $-dE_k/dt$, the enstrophy $\frac{1}{2V} \int \rho |\boldsymbol{\omega}|^2$ and the dilatational fraction, which are the standard diagnostics of three-dimensional compressible test cases [7, 79]; and the Karhunen–Loève spectrum $\sigma_j(t)$. Errors can be measured against a Monte Carlo reference computed with the same deterministic solver on the same n samples: the relative W -Frobenius error $\|\tilde{X} - X\|_W / \|X\|_W$ of the fluctuation matrix, the relative errors of the mean and of the standard deviation fields, and the relative error of the ensemble covariance $\|\tilde{X}\tilde{X}^\top - XX^\top\|_W / \|XX^\top\|_W$. The predictions to test include the defect decomposition (16), the Δt -scaling of Theorem 5.4, the global bound (29) with a posteriori constants, and the cost–error Pareto fronts of Section 6.

9 Numerical experiments

We first verify the theory of Sections 4 to 6 on a stochastic viscous Burgers equation, for which Monte Carlo references and dense evaluations of the right-hand side are inexpensive, so that every quantity entering the analysis can be computed exactly (Section 9.1). We then apply the method to stochastic three-dimensional compressible Navier–Stokes flows (Section 9.2).

9.1 Verification: stochastic viscous Burgers equation

Setup. We solve $\partial_t u + u \partial_x u = \nu \partial_{xx} u$ on $[0, 2\pi)$ with periodic boundary conditions, $\nu = 0.03$, and the random initial condition

$$u(x, 0; \omega) = \sin x + 0.3 \sum_{k=1}^6 \frac{\xi_k(\omega)}{k} \sin(kx + \phi_k(\omega)), \quad \xi_k \sim \mathcal{N}(0, 1), \quad \phi_k \sim \mathcal{U}(0, 2\pi), \quad (45)$$

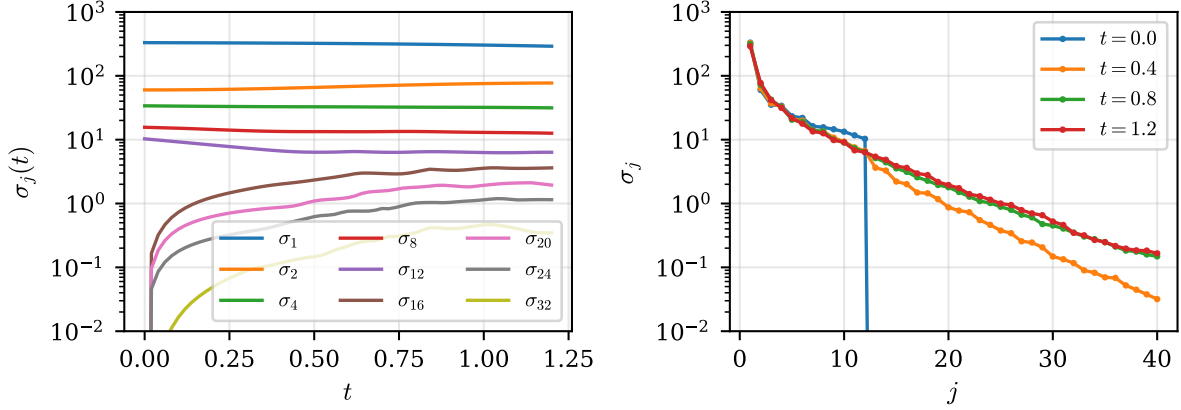


Figure 1: Karhunen–Loève spectrum of the Monte Carlo reference for the stochastic Burgers problem. Left: selected singular values $\sigma_j(t)$; right: the spectrum σ_j at four times. The spectrum broadens as shocks form, so the rank required for a fixed accuracy grows in time.

discretized with $m = 512$ Fourier collocation points, spectral derivatives, 2/3 dealiasing of the product, and the classical fourth-order Runge–Kutta scheme with $\Delta t = 10^{-3}$, which is the stability limit of the explicit treatment of the viscous term ($\nu k_{\max}^2 \Delta t \approx 2$). The stochastic dimension is $d = 12$ and we draw $n = 400$ realizations; the same realizations advanced independently with the same discretization form the Monte Carlo reference $X(t) \in \mathbb{R}^{512 \times 400}$, so that all errors below measure the low-rank approximation alone. The initial matrix has exact rank 12, and its Karhunen–Loève spectrum broadens as the realizations steepen into shocks near $t \approx 1$ (Figure 1): σ_{20} grows from 0.6 to 1.9 and σ_{32} from 0.07 to 0.5 on $[0, 1.2]$, against $\|X\| \approx 330$, so that a rank that is adequate at $t = 0$ is not adequate at $t = 1.2$. The final time is $T = 1.2$.

Implementation. The Runge–Kutta stages are evaluated densely on the low-rank state (no sampling of the right-hand side, so that $\varepsilon_{\text{CUR}} = 0$), the full increment $\Delta t \Phi$ is retracted with two iterations of the deflated robust first-order retraction (24) (Algorithm 1 with the gradient-descent retraction of Charous and Lermusiaux [14, Alg. 2.1] and $N = 2$), and the coefficients are updated by (23). The factorization is rotated to the KL gauge, the scores $s_j = \|P_U^\perp \Phi v_j\|$ (with $v_j = z_j / \sigma_j$, which is well defined for vanishing amplitudes) and $c_j = \|\Phi^\top u_j\|$ are computed, and the sets $\mathcal{A}, \mathcal{B}$ are chosen every $K = 20$ steps ($\tau = 0.02$). Costs are evaluated with the sampled-evaluation column of Table 3 with $q = r + 2$, $p = r$, $c_F = 45$ flops per point (five transforms of length m per column) and two retraction iterations; they are model costs, not timings, because the dense implementation used here to isolate the theory evaluates the right-hand side at all n columns. In these units a Monte Carlo step costs $4nC_m = 3.7 \times 10^7$, a full DORK step 1.6×10^6 at $r = 16$ and 4.9×10^6 at $r = 32$, and the shared cost $C_0(r)$ is 44% ($r = 16$) to 29% ($r = 32$) of the full step, the remainder being the per-mode and per-coefficient work that selective updates reduce.

Local error theorem. Figure 2 tests Theorem 5.4. From KL-gauge rank-16 truncations of the reference at $t = 0.4$ and $t = 1.0$ we take one full step and one selective step with the same Φ for several nested $(\mathcal{A}, \mathcal{B})$ chosen by the sorting rule, and plot $\varrho(\Delta t) = \|\tilde{X}_{i+1}^{\text{sel}} - \tilde{X}_{i+1}^{\text{full}}\| / (\Delta t (\|\varepsilon_{S,i}\|^2 + \|\varepsilon_{Z,i}\|^2)^{1/2})$, the ratio of the measured per-step difference to the leading term of (27). The ratio equals 1 to within 10^{-3} for $\Delta t \leq 10^{-3}$ and its deviation decreases linearly with Δt , as predicted; for constant dynamics

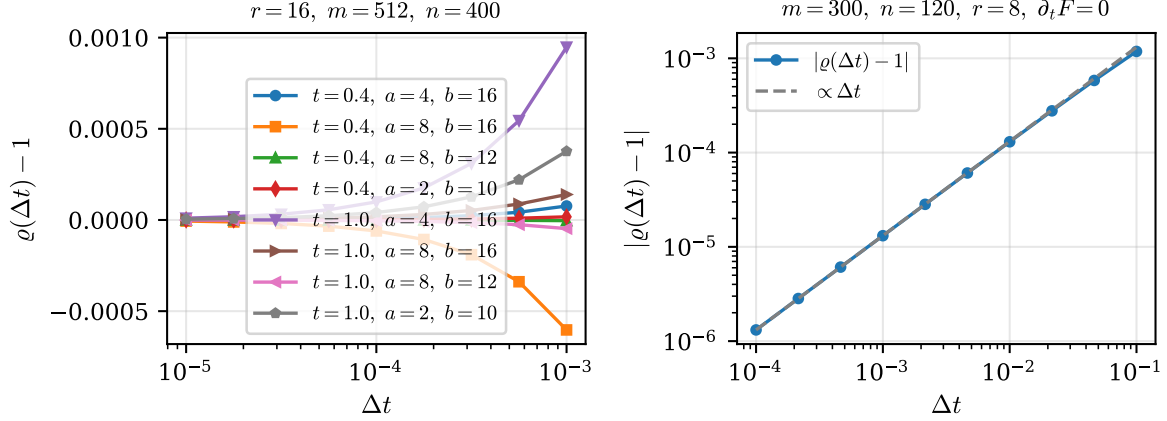


Figure 2: Verification of Theorem 5.4. Left: $\rho(\Delta t) - 1$ for one step from rank-16 KL-gauge truncations of the Burgers reference at $t = 0.4$ and $t = 1.0$ and several (a, b) ; right: $|\rho(\Delta t) - 1|$ for constant dynamics on random matrices, with the reference slope $\propto \Delta t$.

($\partial_t F = 0$, random matrices with $m = 300$, $n = 120$, $r = 8$) the deviation is $|\rho - 1| \approx 10^{-2} \Delta t$ over four decades. The two staleness defects add in quadrature, confirming the orthogonality argument of Appendix A.1.

Selection rule and defect decomposition. Figure 3 compares, at fixed rank $r = 16$ and $\mathcal{B} = [r]$, the full scheme with selective schemes that freeze half of the modes ($a = 8$) according to four rules: the a largest scores s_j (Lemma 4.6), a random subset redrawn at each scheduling instant, the a smallest scores, and the a largest singular values (the energy heuristic). The relative errors at T are 3.6×10^{-2} (full), 3.8×10^{-2} (largest s_j), 4.6×10^{-2} (random), 7.4×10^{-2} (smallest s_j) and 1.1×10^{-1} (largest σ_j); with $a = 4$ they are 4.6, 6.5, 12.9 and 20.2×10^{-2} , and freezing the entire basis ($a = 0$, the POD–Galerkin limit) gives 2.4×10^{-1} . Two conclusions follow. First, updating only the eight modes with the largest s_j costs 6% in error relative to updating all sixteen, whereas updating the eight most energetic modes costs a factor three: the scores $s_j = \sigma_j \|\dot{u}_j\|$ are far from monotone in σ_j , because the weakest modes are the ones that rotate fastest (they live at the small scales that the cascade continuously regenerates), and the score correctly trades amplitude against rotation rate. Second, the ordering of the rules is exactly that of their staleness defects: the left panel of Figure 4 shows $\|\varepsilon_S\|$ for the sorted rule together with $(\sum_{j \notin \mathcal{A}} s_j^2)^{1/2}$, which coincide at every scheduling instant as Theorem 4.4 asserts, and shows that with $a = 8$ the staleness defect is $\approx 19\%$ of the normal defect $\|\varepsilon_{DF}\|$ on average (range 1.8–29 against 13–203), which explains why the error is barely affected; with $a = 0$ the staleness defect exceeds the normal defect. The right panels examine dormant coefficients: freezing the four (respectively eight) coefficients with the smallest c_j in addition to eight bases yields errors of 1.8×10^{-1} and 3.9×10^{-1} , because in this dissipative problem the coefficient velocities $c_j = \|\dot{z}_j\|$ exceed the basis scores by an order of magnitude (the coefficient-staleness defect of the four dormant modes ranges from 4 to 114, against 2–29 for the eight frozen bases): a mode whose amplitude must decay viscously cannot have its amplitude frozen. The scheduler reaches the same conclusion below without being told.

Joint scheduling. Figures 5 to 8 report the scheduled scheme (Algorithm 3) started from the exact rank-12 initial condition with $r_{\max} = 32$, $k_- = 1$, $k_+ = 2$, six geometrically spaced candidate

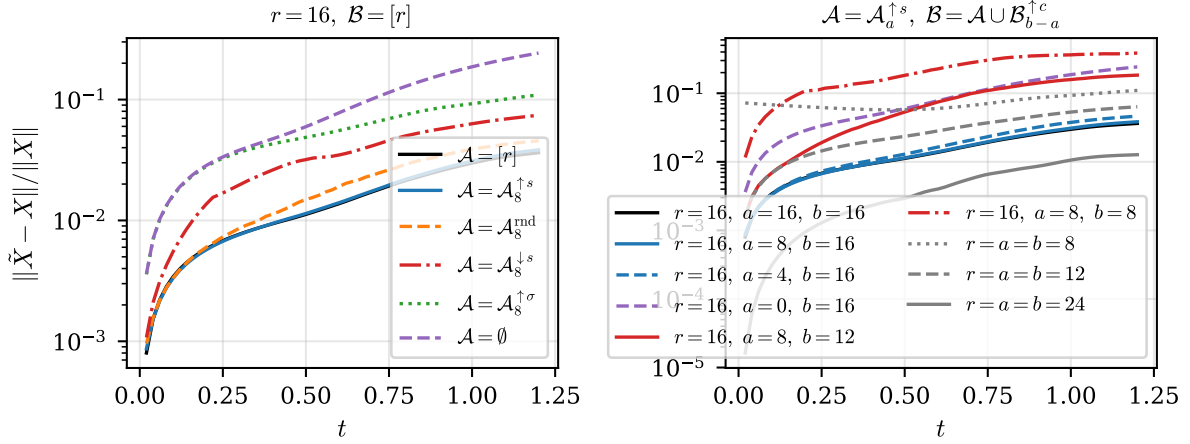


Figure 3: Relative error $\|\tilde{X} - X\| / \|X\|$ versus time for the stochastic Burgers problem at fixed rank. Left: $r = 16, \mathcal{B} = [r]$, and $\mathcal{A}$ chosen as the 8 largest scores s_j ($\mathcal{A}_8^{\uparrow s}$), a random subset ($\mathcal{A}_8^{\text{rnd}}$), the 8 smallest scores ($\mathcal{A}_8^{\downarrow s}$), the 8 largest singular values ($\mathcal{A}_8^{\uparrow \sigma}$), all modes, or no mode. Right: sorted selections with $a \in \{16, 8, 4, 0\}$ and $b = 16$, dormant coefficients ($a = 8, b \in \{12, 8\}$), and full updates at $r \in \{8, 12, 24\}$.

steps in $[10^{-4}, 10^{-3}]$, $\tau_{\max} = 0.02$, a switching cost of two mode updates per tier change, and the error-budget rule (R1) with $\text{tol} \in \{30, 10, 3, 1\}$ (error rate in the Frobenius norm per unit time; $\|X\| \approx 330$). The rank grows from 12 to the cap during shock formation, earlier for tighter tolerances (Figure 5, left). The active count is strongly time dependent: during the smooth phase ($t \lesssim 0.5$) only two to five modes are rotated at $\text{tol} = 30$, essentially the modes just added by the rank increase, which Proposition 5.9(a) forces to be active, while the rest of the basis is frozen at negligible defect; when the shocks form, a rises to 20–30 and later settles around 20 of 32. Averaged over the run, $a/r = 0.42$ at $\text{tol} = 30$ and 0.52 at $\text{tol} = 10$. The scheduler never selected a dormant coefficient ($b = r$ throughout), consistent with the fixed-rank experiment, and the time step stayed at its stability bound 10^{-3} except for one scheduling interval of the $\text{tol} = 3$ run, in which the scheduler selected $\Delta t = 4 \times 10^{-4}$ after a rejected step: in a stability-limited explicit integration the Δt decision is inactive, and the embedded error estimate $\hat{e}/\Delta t \approx 10^{-3}$ is three orders of magnitude below the defects. Once the rank cap is reached the budget becomes infeasible for $\text{tol} \leq 10$ and the fallback of rule (R1) (the cheapest decision within 10% of the minimal achievable error rate) takes over, which is why the $\text{tol} = 3$ and $\text{tol} = 1$ runs differ from $\text{tol} = 10$ only in the number of active modes and not in accuracy.

The resulting work–precision behavior is summarized in Figure 8. To separate the effect of selective updates from that of rank adaptivity, we also ran the scheduler with the same tolerances but with all modes and coefficients forced active ($a = b = r(t)$), i.e., full-update rank-adaptive DORK with the same rank decisions. At $\text{tol} = 30$ the scheduled selective scheme reaches a final relative error of 7.9×10^{-3} at a total model cost of 3.0×10^9 , against 8.4×10^{-3} at 3.8×10^9 for the full-update rank-adaptive scheme (21% cheaper at equal or lower error) and 1.3×10^{-2} at 3.6×10^9 for full updates at fixed $r = 24$; at $\text{tol} = 10$ it reaches 3.9×10^{-3} at 4.0×10^9 , against 4.5×10^{-3} at 4.9×10^9 for the full-update rank-adaptive scheme (19% cheaper) and an estimated 5.9×10^9 for full updates at the fixed rank $r = 32$ that this accuracy requires (best rank-32 approximation error 2.3×10^{-3}). The Monte Carlo reference costs 4.4×10^{10} in the same units, fifteen times the scheduled

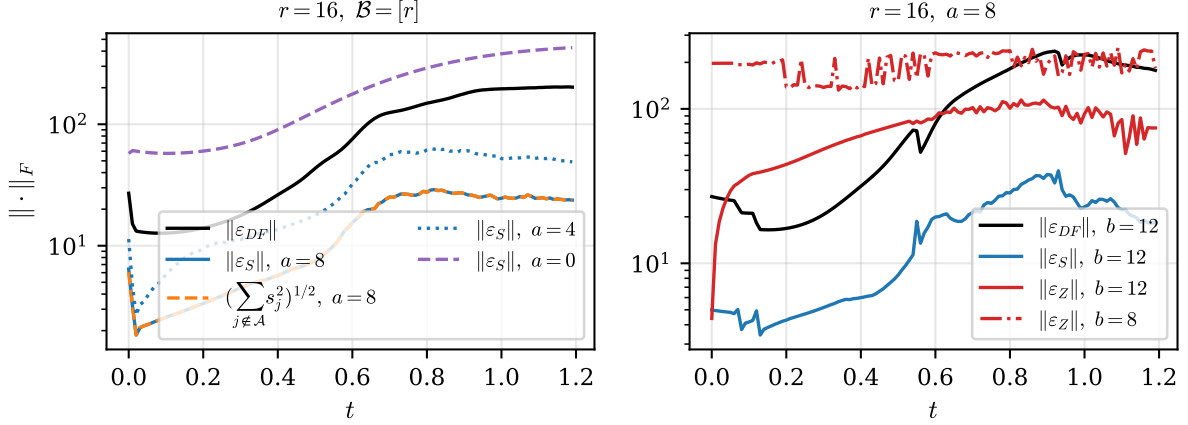


Figure 4: Defect components along the fixed-rank runs of Figure 3 ($r = 16$), evaluated at the scheduling instants. Left: normal defect $\|\varepsilon_{DF}\|$, basis-staleness defect $\|\varepsilon_S\|$ for $a \in \{8, 4, 0\}$, and the score prediction $(\sum_{j \notin \mathcal{A}} s_j^2)^{1/2}$ of Theorem 4.4, which coincides with $\|\varepsilon_S\|$. Right: the three defects with dormant coefficients ($a = 8, b = 12$) and $\|\varepsilon_Z\|$ for $b = 8$.

scheme at $\text{tol} = 30$. The gains come from two sources that the scheduler combines automatically: the rank follows the spectrum (no cost is spent on modes that are not yet needed), and the basis vectors that do not rotate are not rotated; the second source alone accounts for the 20% difference between the selective and full-update rank-adaptive runs, in a problem in which the shared cost $C_0(r)$ that selective updates cannot reduce is 30–45% of the step. Figure 7 shows the Pareto fronts at two instants of the $\text{tol} = 10$ run: at $t = 0.3$ the front spans two orders of magnitude in error rate over a factor of three in cost before the low-gain tail of small time steps, and the budget line selects the corner; at $t = 0.9$ the rank cap makes the budget infeasible and the fallback selects the cheapest near-minimal point. The knee rule (R2), applied in log–log coordinates with a 25% hysteresis band, produces a run of similar cost and accuracy (6.2×10^{-3} at 3.6×10^9) but its decisions alternate between the two corners of the L-shaped front (thin gray line in Figure 5); the switching cost bounds the damage but does not remove the bistability, which is a known weakness of knee selection on fronts with long low-gain tails [8, 21]. We therefore recommend rule (R1) whenever a tolerance is available.

Finally, the right panel of Figure 6 compares the measured error with the time-integrated error rate $\int_0^t (\varepsilon_N + \varepsilon_S) ds$ that the scheduler accumulates, i.e., the bound (28) without the initial-error term and with $\ell = 0$. The integrated rate bounds the error at all times, by a factor of 10–30; the slack is the Gronwall factor with $\ell < 0$ (the neglected small scales are viscously damped, so their defect does not accumulate linearly) and the use of the full residual norm $\|R\|$ as ε_N , both of which the analysis treats conservatively.

9.2 Scope of the compressible-flow assessment

The compressible-flow formulation in Section 8 has not yet been evaluated numerically. The results in this section concern stochastic viscous Burgers flow. A compressible-flow study would require a specified stochastic test case, deterministic discretization, and reference ensemble before its accuracy and cost could be assessed. We therefore report no compressible-flow speedup or validation claim.

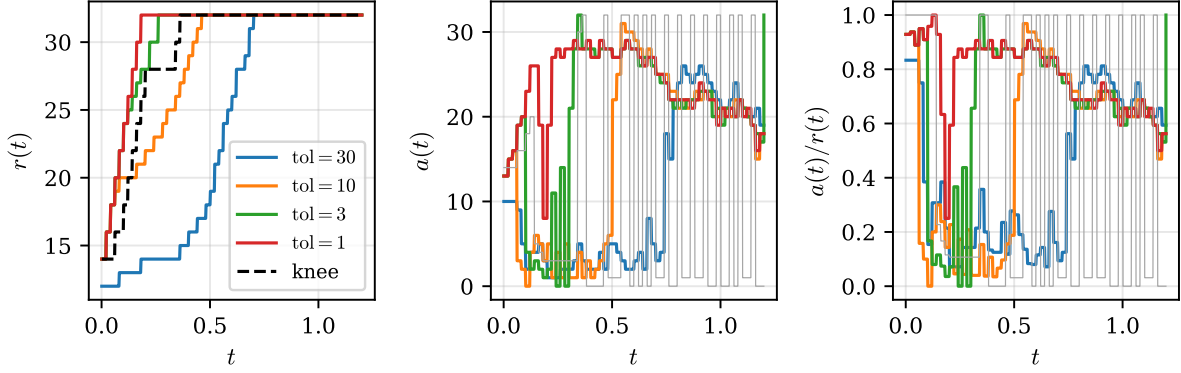


Figure 5: Scheduler decisions along the stochastic Burgers runs under rule (R1) for four tolerances and under the knee rule (R2): rank $r(t)$, number of active bases $a(t)$, and active fraction $a(t)/r(t)$. The number of active coefficients was $b = r$ throughout and the time step remained at its stability bound.

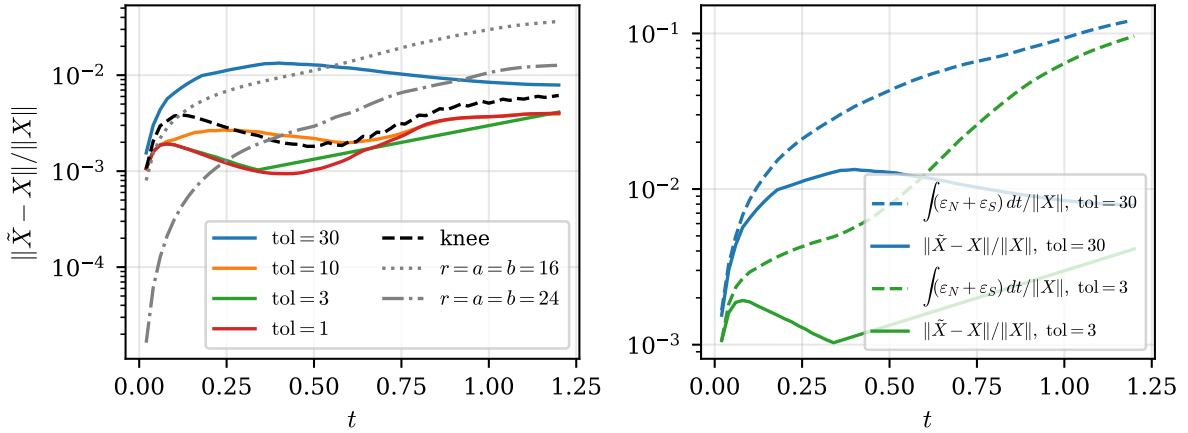


Figure 6: Left: relative error of the scheduled runs of Figure 5 and of the full scheme at fixed rank 16 and 24. Right: measured error and time-integrated scheduled error rate $\int_0^t (\varepsilon_N + \varepsilon_S) ds / \|X\|$ for $\text{tol} = 30$ and $\text{tol} = 3$.

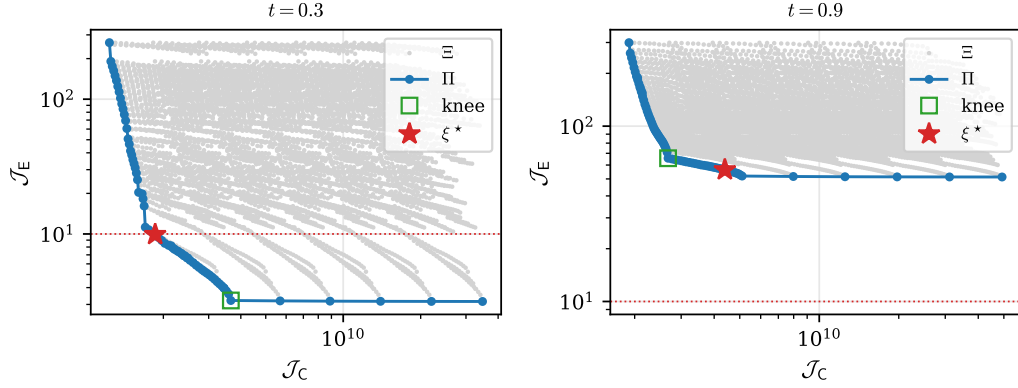


Figure 7: Candidate set Ξ (gray), Pareto front Π (blue), knee (square) and selected decision ξ^* (star) at $t = 0.3$ and $t = 0.9$ of the $\text{tol} = 10$ run; the dotted line is the error budget. At $t = 0.9$ the rank cap makes the budget infeasible and the fallback rule applies.

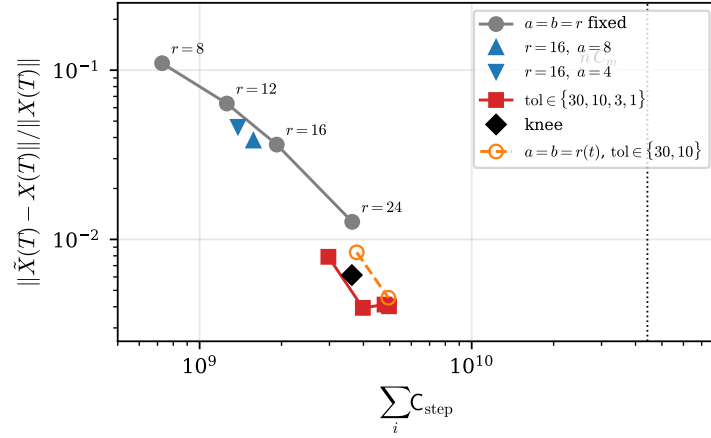


Figure 8: Work–precision diagram for the stochastic Burgers problem: total model cost $\sum_i C_{\text{step}}$ versus final relative error, for the full scheme at fixed rank, the selective scheme at fixed rank 16 with $a \in \{8, 4\}$, the scheduled selective scheme under rule (R1) at four tolerances, under the knee rule, and the full-update rank-adaptive scheme ($a = b = r(t)$) with the same scheduler at $\text{tol} \in \{30, 10\}$; the dotted line marks the Monte Carlo cost.

10 Discussion and conclusions

We introduced rank-adaptive DORK schemes that select which basis vectors and coefficients to advance. A constrained variational principle gives the selective dynamics, and the deflation theorem reduces the active basis update to a standard DORK step (Proposition 4.1 and Theorem 4.7). The exact defect decomposition assigns separate scores to frozen bases and coefficients (Theorem 4.4). Sorting those scores resolves the active-set choice for fixed cardinalities. Local and global error bounds, together with an operation-count model, support joint scheduling of the time step, rank, and active sets.

Verification and limitations. The stochastic Burgers experiments verify the defect decomposition and local error behavior. In this example, selecting modes by their defect scores improves on both random selection and an energy-based rule. The observed cost comparison depends on the stated operation-count model and test configuration. Discrete bounds can depend on small active singular values when nonrobust retractions are used [14, 48]. The scheduler makes decisions over individual intervals, and the formulation does not enforce positivity or conservation for every reconstructed compressible realization. These properties require further numerical and analytical study.

Further work. A closure could predict the contributions of frozen factors from their recent history, provided its residual is estimated and incorporated into the error budget. Robust retractions, longer-horizon scheduling, and selective sampling of realizations are additional directions. The compressible Navier–Stokes formulation in Section 8 establishes the equations and cost model; its numerical results remain to be completed.

A Proofs

A.1 Proof of Theorem 5.4

Drop the step index and write $\tilde{X} = UZ^\top$, $\chi = \tilde{X} + \Delta t\Phi$, $\dot{U} = P_U^\perp \Phi Z(Z^\top Z)^{-1}$ (full first-order correction, (9)) and $\dot{U}_A = P_U^\perp \Phi Z_A(Z_A^\top Z_A)^{-1}$ (selective first-order correction, (22)). All retractions of the family of Charous and Lermusiaux [14] produce a new basis of the form $U^f = \text{orth}(U + \Delta t\dot{U} + \Delta t^2 V_2^f + \dots)$ with $U^\top \dot{U} = 0$ and, by Theorem 4.7, $U^s = [U_{\mathcal{F}} \text{ orth}_{\perp U_{\mathcal{F}}}(U_A + \Delta t\dot{U}_A + \Delta t^2 V_2^s + \dots)]$, where the second-order corrections satisfy $\|V_2^f\|, \|V_2^s\| \leq \kappa_2 \|\Phi\|^2 / \sigma_{\min}^2$ [14, Eq. (2.14b)], and both produce the optimal coefficients $Z^f = \chi^\top U^f$, $Z_B^s = \chi^\top U_B^s$, $Z_D^s = Z_D$. Hence

$$\tilde{X}^{\text{sel}} - \tilde{X}^{\text{full}} = P_{U_B^s} \chi + U_D Z_D^\top - P_{U^f} \chi = (P_{U^s} - P_{U^f}) \chi + U_D (Z_D^\top - U_D^\top \chi), \quad (46)$$

using $P_{U^s} = P_{U_B^s} + P_{U_D}$ ($\mathcal{D} \subseteq \mathcal{F}$, so $U_D^s = U_D$). The last term equals $-\Delta t U_D U_D^\top \Phi = -\Delta t \varepsilon_Z$ exactly, because $U_D^\top \chi = Z_D^\top + \Delta t U_D^\top \Phi$.

For the projector difference, let $W(\Delta t) = \text{orth}(W_0 + \Delta t V + \Delta t^2 V_2 + \dots)$ with $W_0^\top W_0 = I$ and $W_0^\top V = 0$. The polar factor $(W_0 + \Delta t V + \Delta t^2 V_2)(I + \Delta t^2 V^\top V + \Delta t^2 (W_0^\top V_2 + V_2^\top W_0) + \dots)^{-1/2}$ gives

$$P_{W(\Delta t)} = P_{W_0} + \Delta t (V W_0^\top + W_0 V^\top) + E, \quad \|E\| \leq \Delta t^2 (2\|V\|^2 + 2\|V_2\|) + \mathcal{O}(\Delta t^3), \quad (47)$$

since the projector is invariant under the choice of orthonormalization within the span. Apply (47) to U^f ($W_0 = U$, $V = \dot{U}$) and to the active block of U^s ($W_0 = U_A$, $V = \dot{U}_A$, with $P_{U^s} = P_{U_{\mathcal{F}}} + P_{\text{active block}}$ because the two blocks stay orthogonal):

$$(P_{U^s} - P_{U^f}) \chi = \Delta t [(\dot{U}_A U_A^\top + U_A \dot{U}_A^\top) - (\dot{U} U^\top + U \dot{U}^\top)] \chi + (E^s - E^f) \chi.$$

Now $\dot{U}U^\top\chi = \dot{U}Z^\top + \Delta t\dot{U}U^\top\Phi = P_U^\perp\Phi P_Z + \Delta t\dot{U}U^\top\Phi$ and $U\dot{U}^\top\chi = \Delta tU\dot{U}^\top\Phi$ because $\dot{U}^\top U = 0$ annihilates $\dot{U}^\top UZ^\top$; likewise $\dot{U}_\mathcal{A}U_\mathcal{A}^\top\chi = P_U^\perp\Phi P_{Z_\mathcal{A}} + \Delta t\dot{U}_\mathcal{A}U_\mathcal{A}^\top\Phi$ and $U_\mathcal{A}\dot{U}_\mathcal{A}^\top\chi = \Delta tU_\mathcal{A}\dot{U}_\mathcal{A}^\top\Phi$. Therefore

$$(P_{U^s} - P_{U^f})\chi = -\Delta t P_U^\perp\Phi(P_Z - P_{Z_\mathcal{A}}) + \mathcal{O}(\Delta t^2) = -\Delta t \varepsilon_S + \mathcal{O}(\Delta t^2),$$

which together with (46) proves the first identity in (27). The $\mathcal{O}(\Delta t^2)$ remainder is bounded by

$$\Delta t^2 \left[2(\|\dot{U}\| + \|\dot{U}_\mathcal{A}\|) \|\Phi\| + (2\|\dot{U}\|^2 + 2\|\dot{U}_\mathcal{A}\|^2 + 2\|V_2^f\| + 2\|V_2^s\|) \|\chi\| \right] + \mathcal{O}(\Delta t^3),$$

and $\|\dot{U}\| \leq \|\Phi\| \|Z(Z^\top Z)^{-1}\| = \|\Phi\| / \sigma_{\min}$, $\|\dot{U}_\mathcal{A}\| \leq \|\Phi\| / \sigma_{\min, \mathcal{A}} \leq \|\Phi\| / \sigma_{\min}$ (Cauchy interlacing, Proposition 5.9(c)), which gives the stated form of C_i . Finally, ε_S has its column space in range $U^\perp$ and ε_Z in range $U_\mathcal{D}$, so $\|\varepsilon_S + \varepsilon_Z\|^2 = \|\varepsilon_S\|^2 + \|\varepsilon_Z\|^2$, and the triangle inequality yields the norm bound. Since every retraction of the family agrees with $\tilde{X} + \Delta t P_{T_{\tilde{X}} \mathcal{M}_r} \Phi$ to first order [14, Table 2.2], the leading term does not depend on the retraction order. $\square$

A.2 Proof of Theorem 5.5

(i) Let $e(t) = \tilde{X}(t) - X(t)$. On each scheduling interval the sets $(\mathcal{A}, \mathcal{B}, r)$ are constant and $\tilde{X}$ solves $\dot{\tilde{X}} = P_{\tilde{X}}^{\mathcal{A}, \mathcal{B}} F(\tilde{X}, t)$, so

$$\dot{e} = (F(\tilde{X}, t) - F(X, t)) - (F(\tilde{X}, t) - P_{\tilde{X}}^{\mathcal{A}, \mathcal{B}} F(\tilde{X}, t)).$$

Taking the inner product with e , using the one-sided Lipschitz condition of Assumption 5.1, the Cauchy–Schwarz inequality and Theorem 4.4 with Assumption 5.2,

$$\frac{1}{2} \frac{d}{dt} \|e\|^2 \leq \ell \|e\|^2 + \|e\| (\|\varepsilon_{DF}\|^2 + \|\varepsilon_S\|^2 + \|\varepsilon_Z\|^2)^{1/2} \leq \ell \|e\|^2 + \|e\| (\varepsilon_N + \varepsilon_S + \varepsilon_Z),$$

hence $\frac{d}{dt} \|e\| \leq \ell \|e\| + \varepsilon_N + \varepsilon_S + \varepsilon_Z$ in the sense of Dini derivatives. The state $\tilde{X}$ is continuous across scheduling instants (a tier change alters the velocity, not the state), so $\|e\|$ is continuous and piecewise differentiable on $[0, t]$, and Gronwall’s inequality [36, 40] gives (28).

(ii) Let $\varphi_{s \rightarrow t}$ denote the flow of the selective DO equations with the prescribed schedule, and $\hat{y}_i = \varphi_{0 \rightarrow t_i}(\tilde{X}_0)$, so that $\hat{y}_N = \varphi_{0 \rightarrow T}(\tilde{X}_0)$ is bounded by (i): $\|\hat{y}_N - X(T)\| \leq e^{\ell T} \|\tilde{X}_0 - X(0)\| + \sum_i \Delta t_i \omega_i (\varepsilon_{N,i} + \varepsilon_{S,i} + \varepsilon_{Z,i})$ with $\omega_i = \Delta t_i^{-1} \int_{t_i}^{t_{i+1}} e^{\ell(T-s)} ds \leq e^{\ell(T-t_{i+1})} \max(1, e^{\ell \Delta t_i})$, by bounding the integrand on each interval by its supremum. Write $\mathcal{S}^{(i \rightarrow N)} = \mathcal{S}_{N-1} \circ \dots \circ \mathcal{S}_i$ (identity for $i = N$) and telescope,

$$\tilde{X}_N - \hat{y}_N = \sum_{i=0}^{N-1} \left[\mathcal{S}^{(i+1 \rightarrow N)}(\mathcal{S}_i(\hat{y}_i)) - \mathcal{S}^{(i+1 \rightarrow N)}(\varphi_{t_i \rightarrow t_{i+1}}(\hat{y}_i)) \right].$$

By Assumption 5.3, each bracket is bounded by $\prod_{k=i+1}^{N-1} (1 + L_S \Delta t_k) (C_{RK} \Delta t_i^{p+1} + C_{ret} \Delta t_i^{q+1}) \leq e^{L_S(T-t_{i+1})} (C_{RK} \Delta t_i^{p+1} + C_{ret} \Delta t_i^{q+1})$, the “Lady Windermere’s fan” estimate [40, Sec. II.3]. Adding the two contributions gives (29). $\square$

A.3 Derivation of the drift bounds (32)

Work in the KL gauge at time t , so $\dot{u}_j = P_U^\perp F z_j / \sigma_j^2$, $\dot{z}_j = F^\top u_j$, $\|\dot{U}\| \leq \|F\| / \sigma_{\min}$ and $\|\dot{Z}\| = \|F^\top U\| \leq \|F\|$. Write $w_j = P_U^\perp F z_j$, so that $s_j = \|w_j\| / \sigma_j$. Along the selective (or full) flow,

$$\dot{w}_j = -(\dot{U}U^\top + U\dot{U}^\top) F z_j + P_U^\perp \dot{F} z_j + P_U^\perp F \dot{z}_j, \quad \|\dot{F}\| \leq \ell_+ \|\dot{X}\| + \|\partial_t F\| \leq \ell_+ \|F\| + \|\partial_t F\|,$$

where we used $\|\tilde{X}\| = \|P_{\tilde{X}}^{\mathcal{A},\mathcal{B}} F\| \leq \|F\|$. Hence $\|\dot{w}_j\| \leq 2 \|\dot{U}\| \|F\| \sigma_j + (\ell_+ \|F\| + \|\partial_t F\|) \sigma_j + \|F\|^2 \leq \sigma_j (2F_{\max}^2/\sigma_{\min} + \ell_+ F_{\max} + F'_{\max}) + F_{\max}^2$. The singular value σ_j is the square root of the j th eigenvalue of $Z^\top Z$; by first-order eigenvalue perturbation, $\frac{d}{dt} \sigma_j^2 = 2z_j^\top \dot{z}_j = 2z_j^\top F^\top u_j$, so $|\dot{\sigma}_j| \leq \|F^\top u_j\| \leq F_{\max}$. Therefore

$$|\dot{s}_j| \leq \frac{\|\dot{w}_j\|}{\sigma_j} + \frac{\|w_j\| |\dot{\sigma}_j|}{\sigma_j^2} \leq \frac{2F_{\max}^2}{\sigma_{\min}} + \ell_+ F_{\max} + F'_{\max} + \frac{F_{\max}^2}{\sigma_j} + s_j \frac{F_{\max}}{\sigma_j} \leq \ell_+ F_{\max} + F'_{\max} + \frac{4F_{\max}^2}{\sigma_{\min}},$$

using $s_j = \|P_U^\perp F z_j\| / \sigma_j \leq \|F\|$. For the coefficient scores, $c_j = \|F^\top u_j\|$ and $|\dot{c}_j| \leq \|\dot{F}\| + \|F\| \|\dot{u}_j\| \leq \ell_+ F_{\max} + F'_{\max} + F_{\max}^2 / \sigma_j$. The Euclidean norms of the r -vectors $\dot{s}, \dot{c}$ are bounded by $\sqrt{r}$ times the componentwise bounds, which gives (32). The discrete scores computed with Φ in place of F differ from these by $\mathcal{O}(\Delta t)$. $\square$

B Notation

Table 4: Symbols used throughout the paper.

Symbol	Meaning
$X(t) \in \mathbb{R}^{m \times n}$	full-rank state (m spatial degrees of freedom, n realizations)
$\tilde{X} = UZ^\top \in \mathcal{M}_r$	rank- r approximation; $U \in \mathcal{V}_{m,r}$ orthonormal modes, $Z \in \mathbb{R}_*^{n \times r}$ coefficients
$F(X, t), \Phi$	right-hand side of (1); integrated operator over one step, (6)
$\chi = \tilde{X}_i + \Delta t \Phi$	before retraction point
$\mathcal{M}_r, T_{\tilde{X}} \mathcal{M}_r, P_{T_{\tilde{X}} \mathcal{M}_r}$	rank- r manifold, tangent space, tangent projector (3)
$P_U, P_U^\perp, P_Z, P_Z^\perp$	projectors onto/orthogonal to range U and range Z
σ_j, Λ , KL gauge	singular values of $\tilde{X}$; $Z^\top Z = \Lambda = \text{diag}(\sigma_j^2)$
$\dot{u}_j = P_U^\perp F z_j / \sigma_j^2, \dot{z}_j = F^\top u_j$	DO mode and coefficient velocities (KL gauge)
$\mathcal{A}, \mathcal{F} = [r] \setminus \mathcal{A}; a = \mathcal{A} $	active / frozen basis sets
$\mathcal{B}, \mathcal{D} = [r] \setminus \mathcal{B}; b = \mathcal{B} $	active / dormant coefficient sets
$\mathcal{M}_r^{\mathcal{F}}, \mathcal{S}_{\tilde{X}}^{\mathcal{A},\mathcal{B}}, P_{\tilde{X}}^{\mathcal{A},\mathcal{B}}$	constrained manifold (10), admissible velocities (12), selective projector (14)
$s_j = \sigma_j \ \dot{u}_j\ , c_j = \ \dot{z}_j\ $	basis and coefficient staleness scores (17)
$\varepsilon_{DF}, \varepsilon_S, \varepsilon_Z$	normal, basis-staleness, coefficient-staleness defects (Table 2)
$\varepsilon_N(t), \varepsilon_S(t), \varepsilon_Z(t)$	scheduled bounds on the defects (Assumption 5.2)
$\chi' = P_{U_{\mathcal{F}}}^\perp \chi, \Phi' = P_{U_{\mathcal{F}}}^\perp \Phi$	deflated point and increment (Theorem 4.7)
$\mathcal{R}_{\tilde{X}}(\Delta t \Phi)$	retraction; q its order
$p, \hat{e}, \Delta$	Runge–Kutta order, embedded error estimate, retraction estimate
ℓ, ℓ_+, L_S	one-sided/two-sided Lipschitz constants of F ; Lipschitz constant of the one-step map
L_s, L_c	drift constants of the score vectors (Lemma 5.7)
$t_k, \tau_k = K_k \Delta t_k$	scheduling instants and interval lengths
$C_0, C_U, C_Z, C_{\text{sch}}, C_{\text{sw}}$	cost-model coefficients (34)
$\rho, \mathcal{I}_C, \mathcal{I}_E, \Xi, \Pi$	error rate (30), objectives (40), decision set, Pareto front
q, p (sampling), $\mathcal{J}, \mathcal{I}$	numbers and index sets of sampled realizations and spatial points (44)
W	weighted inner product for compressible variables (Section 8.2)

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